

**An internship report
On**

“Power Electronic Circuits - Numerical Simulations”

Submitted in partial fulfilment of the requirement of Master of Technology
SRM University, Amaravathi, AP



Electronics and Communication Engineering

2023 – 2025

Under the Guidance of

Dr. Naga Brahmendra Yadav Gorla Ph.D
Assistant Professor

Submitted by

Jagadeeswara Chandra bose. K
M. Tech _AP23122100002

Summer Internship Carried out at



IIT PALAKKAD

“Indian Institute of Technology PALAKKAD”

Kanjikode, Palakkad Kerala 678623.

Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the summer internship report titled “**Power Electronic Circuits - Numerical Simulations**” is a bonafide work carried out by **JAGADEESWARA CHANDRA BOSE. K** bearing Redg no: AP23122100002 in partial fulfilment of the requirements for the summer internship for the award of degree of Master of Technology in Electronics and Communication Engineering of SRM University, Amaravathi, AP during the academic year 2023-2025. The summer internship report has been approved as it satisfies the academic requirements prescribed by the university.

 12/07

Signature of the Guide

Dr. Naga Brahmendra Yadav Gorla Ph.D
Assistant Professor
Dept of EE
Indian Institute of Technology OF PALAKKAD,
KERALA

Signature of the HOD

Dr. Ramesh Vaddi Ph.D
Head of the Department,
Dept of ECE
SRM UNIVERSITY, AP

DECLARATION

I, Jagadeeswara chandra bose. K[Regd: AP23122100002], student of III Semester M. Tech , in Electronics and Communication Engineering, SRM University hereby declare that the Internship entitled “Power Electronic Circuits - Numerical Simulations” has been carried out by me at Indian Institute of Technology of Palakkad and submitted in partial fulfilment of the requirements of the summer internship for the award of degree of Master of Technology in Electronics and Communication Engineering of SRM University, AP during academic year 2023-2025.

Date: 16/07/2024.

Place: Nerukonda, Amaravathi, AP

Name: Jagadeeswara Chandra bose
SRM, AP Regd: AP23122100002

Jagadeeswara chandra bose. K
Electronics and Communication Engineering
Master of Technology, Embedded Systems and IoT
School of Engineering and Sciences
SRM University AP

I am pleased to submit my project report documenting the work I completed during my summer internship in the Electrical Engineering Department at IIT Palakkad. Under your guidance of **Dr. Naga Brahmendra Yadav Gorla Ph.D, Assistant Professor**, I had the opportunity to participate in the research project "**Power Electronic Circuits - Numerical Simulations.**"

This internship provided me with valuable hands-on experience in numerical simulations of power electronic circuits. I gained proficiency in using circuit simulation software to model and analyze the behavior of various power electronic topologies. Through this project, I developed a deeper understanding of the design considerations and trade-offs involved in power converter circuits.

In the attached report, I have documented the objectives, methodology, and key findings of my internship work. The report includes detailed circuit simulations, performance analyses, and design optimizations for several power electronic circuits. I believe the insights gained from this project will be beneficial for future research and development in this field.

I would like to express my sincere gratitude for your mentorship and support throughout this internship. Your expertise and guidance were instrumental in the successful completion of this project. I am grateful for the opportunity to have worked with you and the Electrical Engineering team at IIT Palakkad.

Please find the project report attached for your review. I am available to discuss the report or any aspects of my internship experience in more detail.

Thank you for considering my work.

Best Regards,





INDIAN INSTITUTE
OF TECHNOLOGY
PALAKKAD

Summer Internship 2024

**Indian Institute of Technology
Palakkad**

This is to certify that Mr. Kamarajugadda Jagadeeswara Chandra Bose has worked in the area Power Electronic Circuits - Numerical Simulations as a Summer Research Fellow under the supervision of Dr. Naga Brahmendra.



Place: Palakkad
Date: 16-07-2024

Lakshmi
Associate Dean (Academics, PG)



IITPKD/ACAD/SI/OFLT/2024/25

03-05-2024

Provisional Offer of admission to the Summer Internship Program 2024

Ref.: Your application for admission to the Summer Internship Program 2024 of IIT Palakkad
(Application No. 2024SUM3909)

Dear Jagadeeswara chandra bose,

I am glad to inform you that you have been provisionally selected for admission to the Summer Internship Program 2024 in the Department of Electrical Engineering. You are selected to work under the supervision of Dr. Naga Brahmendra. Given below are the terms and conditions of the internship:

1. Your acceptance of the internship has to be intimated by email to summer-internship@iitpkd.ac.in. You also need to fill this google form ([Click here](#)) and update the details regarding acceptance/non-acceptance of the offer. If it is not received before 08th of May 2024, it will be assumed that you are not planning to take it up, and the internship will be offered to a person from the waiting list.
2. The Internship duration is from 03rd of June 2024 till 16th of July 2024. You will have to stay in the campus during this entire period excluding holidays. On satisfactory completion, you will be issued a certificate of participation in the program.
3. Under the program, you will be eligible for a financial support amounting to Rs.8,000/- per month (Rs.12,000/- for the entire period).
4. The certificate and the financial support will be given only if you stay for the entire duration of the internship.
5. Accommodation on a shared basis is available on campus. The rooms have only cots. You will have to bring your own mattresses, bed sheets and pillows. Food will be available in the mess. Please note that you will be charged a hostel room rent of Rs.100/- per day and mess food charges of Rs.160/- per day.
6. You are advised to report on **03rd of June 2024** between 9 am and 10 am, for getting the room allotment. **No allotment will be made prior to the date mentioned above.** In case of you reaching IIT Palakkad prior to 03-06-2024, you need to make your own arrangements for accommodation. You may plan your travel accordingly.
7. You have to produce the originals of relevant certificates including the original bonafide certificate issued by your institute with one set of photocopies.

Yours sincerely,

Dr. Lakshmi Sankar K
Associate Dean(Academics, PG)
IIT Palakkad

Copy to:

1. Dr. Naga Brahmendra
2. Joint Registrar (Academics)
3. Students Section

Acknowledgment

I would like to express my sincere thanks to everyone who supported and guided me during my summer internship at the Indian Institute of Technology Palakkad.

First, I would like to thank my project supervisor, Professor **Dr. Naga Brahmendra Yadav Gorla Ph.D**, Assistant Professor from IIT Palakkad, for giving me this opportunity to work on the "Power Electronic Circuits - Numerical Simulations" project. His expertise, guidance, and encouragement were crucial for the successful completion of this internship.

I am grateful to the Electrical Engineering Department at IIT Palakkad for their support and for providing me with the necessary resources and facilities for my project work.

I would also like to acknowledge the help and guidance provided by the research scholars and lab staff in the department. Their insights and suggestions were valuable in overcoming challenges during the project.

I would like to thank my college faculty, **Dr. Ramakrishnan Maharajan**, **Dr. Ramesh vaddi** for allowing me for this internship opportunity and for their continuous support throughout my academic journey.

Finally, I would like to thank my family and friends for their unwavering support and encouragement during my internship experience. Their belief in me was a constant source of motivation.

This internship has been a transformative experience, and I am grateful to have worked with such a talented and dedicated team at IIT Palakkad. The knowledge and skills I gained during this period will be valuable assets in my future endeavors.

Abstract:

Power electronic circuits are widely used in various applications such as power conversion, motor drives, renewable energy systems, and power supplies. Numerical simulations play a crucial role in the design, analysis, and optimization of these circuits. This report presents an overview of the key aspects involved in the numerical simulation of power electronic circuits. The fundamental principles of power electronics, including switching devices, converter topologies, and control strategies, are discussed. The importance of accurate modeling and simulation techniques for predicting the behavior of power electronic circuits is emphasized. The report also highlights the challenges and limitations associated with numerical simulations and provides insights into the future trends and research directions in this field. The presented information can serve as a valuable resource for researchers, engineers, and students working in the area of power electronics and its applications.

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Plan of Internship program

Research Summer Internship IIT Palakkad - 45days [6 weeks 3/6/2024 - 16/7/2024]			
Topic	Power Electronic Circuits - Numerical Simulations	Dept	EE - Electrical Engineering
		Faculty	Dr. Naga Brahendra - Assistant Professor
Weeks	Details	Weeks	Details
Week - 1	Introduction to power electronics. Software setup[LT spice, Altium Pcb design]. Voltage pulse generator and MOSFET charactries. Power electronics converters _circuit practice[12 Examples] [Learnings - Converters and EDA Tools]	Week - 4	NCP 51820 Gate Driver _ ON semi GaN 3R2_100 CBE _ MOSFET _ Nexperia Data sheet, Symbol and Component creation in LT Spice and Altium PCB design [learnings - Creating of Libraries in EDA Tools and Gate driver and MOSFET]
Week - 2	Power Electronics Circuits design and Numerical calculations and creating a formula sheet and Simulating in LT spice tool. 1.Buck 2.Boost and 3.Buck Boost [Learnings - Calculation and circuit design and simulations in LTspice]	Week - 5	Design and Simulating of Synchronous Buck boost converter Schmetic design and PCB layout component assembly and routing [Learnings - Altium PCB desining]
Week - 3	Power Electronics Circuits design and Numerical calculations and creating a formula sheet and Simulating in LT spice tool. 1.Forward [CCM & DCM] 2. Flyback 3.Pushpull [Learnings - Calculation and circuit design and simulations in LTspice]	Week - 6	

Out comes

- * Power Electronics Converters
- * Calcuation and formula based sheets creation
- * Circuit Simulation in LT Spice Tool
- * Gate Drivers _ NCP51820 _ Onsemi
- * GaN MOSFET GAN3R2-100CBE _ Nexperia
- * EDA Tools - Altium PCB design
- * Skills - LT spice simulations, Numerical calculations and Altium Schematic and PCB designing.

Overview of IIT Palakkad

IIT Palakkad is a public autonomous engineering and research institute in Palakkad, Kerala, India. It is one of the 23 Indian Institutes of Technology (IITs) and was established in 2015 as an Institute of National Importance by the Government of India.

Rankings and Recognition

IIT Palakkad ranked 69th under the 'Engineering' category by the National Institutional Ranking Framework (NIRF) Ranking 2023.

Vision

Purpose

To create knowledge; to communicate knowledge; and to apply knowledge for the benefit of society.

To nurture lifelong learners who have a rigorous foundation, a passion for innovation, strong ethical values and a recognition of their social responsibility.

To be home to the best minds from across the globe.

Core Values

We are a happy and vibrant community founded on reason and compassion, driven by curiosity, creativity and a desire to be agents of change.

We cherish a sense of belonging towards the Institution.

We foster diversity in composition and thought, boundaryless inquiries and interactions, and free and independent thinking.

We espouse an ecologically and socially responsible way of life.

Envisioned Future

We envision being a globally recognized institution at the forefront of education, research, and technology development, having strong ties with industries and other academic institutions.

We envision being an inclusive and diverse community on a sustainable and green campus, actively engaged with our neighbourhood.

Objectives and Scope of the Project

Objectives:

- Power Electronics Converters: Design and analysing power electronics converter links.
- Converters Analysis: Numerical calculations and analysis for various converters including buck, boost, and buck-boost converters, flyback, forward, and push-pull converters.
- Spreadsheet Development: Create a comprehensive formula-based spreadsheet for converter calculations.
- Simulation: Simulate the converters using LT Spice to validate designs and calculations.
- Gate Driver Circuit Design: Design the NCP51820 Onsemi gate driver circuit.
- Component Creation: Develop and integrate GaN 3R2-100CBE components.
- Schematic and PCB Design: Created detailed Altium schematics and PCB routing.

Scope:

- Research and Development: Focus on advancing the design and efficiency of power electronic converters.
- Practical Applications: Apply theoretical knowledge to practical designs and simulations.
- Tool Proficiency: Gain expertise in tools like LT Spice and Altium for simulation and schematic design.
- Component Integration: Learn the process of creating and integrating advanced components such as GaN transistors and Gate drivers into circuit designs.

Chapter – 1

Power Electronics

1.1. Introduction to Power Electronics

Power electronics is a crucial field in electrical engineering, focusing on the efficient conversion and control of electrical power using electronic devices. It encompasses the design, analysis, and implementation of circuits that convert electrical energy from one form to another, which is essential for various applications ranging from consumer electronics to industrial systems and renewable energy sources.

The field of power electronics has evolved significantly over the years, driven by advancements in semiconductor technology, which have enabled the development of more efficient, reliable, and compact power converters. These converters include rectifiers, inverters, choppers, and various types of DC-DC converters such as buck, boost, and buck-boost converters.

Power electronics plays a pivotal role in modern technology, facilitating the control of power flow and enhancing the performance of electrical systems. It enables efficient energy conversion in a wide range of applications, including power supplies for electronic devices, electric vehicles, renewable energy systems, and industrial automation.

The importance of power electronics lies in its ability to improve energy efficiency, reduce losses, and provide precise control over electrical power. As energy demands continue to rise and the push for renewable energy sources intensifies, the role of power electronics in developing sustainable and efficient energy solutions becomes increasingly vital.

1.2. Power Electronics Converters

Buck Converter:

A buck converter is a type of DC-DC converter that steps down the input voltage to a lower output voltage. It is widely used in applications where a stable and lower voltage is needed from a higher voltage source, such as in power supplies for electronic devices. The buck converter operates by switching a transistor on and off, storing energy in an inductor, and releasing it at a controlled rate to the load.

Boost Converter:

A boost converter is a DC-DC converter that steps up the input voltage to a higher output voltage. It is useful in applications where an increase in voltage is required, such as in battery-powered devices needing a higher operating voltage than the battery provides. The boost converter works by storing energy in an inductor when the switch is closed and releasing it to the load through a diode when the switch is open.

Buck-Boost Converter:

A buck-boost converter combines the functionalities of both buck and boost converters, allowing it to either step up or step down the input voltage based on the requirements. It is versatile and used in applications where the output voltage needs to be adjusted above or below the input voltage. The buck-boost converter can invert the voltage, making it useful in a variety of power management scenarios.

Flyback Converter:

The flyback converter is a type of DC-DC converter that uses a transformer to store energy when the switch is on and transfer it to the output when the switch is off. It can provide isolation between the input and output, which is crucial in applications requiring safety and noise reduction. Flyback converters are commonly used in low-power applications such as power supplies for televisions and other consumer electronics.

Forward Converter:

A forward converter is another type of isolated DC-DC converter that uses a transformer for energy transfer. Unlike the flyback converter, the forward converter transfers energy to the output during the switch's on-state. It is typically used in higher power applications and offers better efficiency and regulation compared to the flyback converter. Forward converters are found in industrial power supplies and telecom equipment.

Chapter - 2

Software Setup and Tools

2.1. LTspice Simulation Software

LTspice: A Powerful, Free SPICE Simulator

LTspice is a powerful, fast, and free SPICE simulation software created by Linear Technology Company, now known as Analog Devices. It encompasses a graphical schematic capture interface that lets users to probe the integrity of specific circuits and the creation, of course, of simulation results according to the settings of simulation that the user will see through a built-in waveform viewer, through the selection of different modes and methodologies that are aimed to investigate the electrical circuits in question.

Key Features and Benefits of LTspice

Fast and Unlimited Simulation Capabilities: LTspice outperforms many paid SPICE simulation options, with an unlimited number of nodes and the ability to rapidly simulate complex analog circuits.

Extensive Model Library: LTspice comes with over 1,100 macromodels for Analog Devices products, as well as a library of passive devices to aid circuit design.

Optimized for Power Electronics: LTspice is particularly well-suited for simulating switching mode power supplies (SMPS), with features like steady state detection and efficiency/power computations.

Free and Widely Used: As a free SPICE simulator, LTspice is widely adopted by electronics engineers and students. It provides a cost-effective solution for circuit prototyping and analysis.

Advanced LTspice Capabilities

Beyond the basic simulation capabilities, LTspice offers several advanced features:

Importing Third-Party Models: LTspice allows users to incorporate external SPICE models, enabling the simulation of a wider range of components.

Behavioral Sources and Expressions: LTspice supports the use of behavioral sources and the evaluation of mathematical expressions, providing greater flexibility in circuit modeling.

Temperature Analysis: LTspice enables the simulation of circuits under varying temperature conditions, allowing designers to assess the impact of temperature on circuit performance.

2.2. Altium Designer for PCB Design

Developer : Altium Limited

Initial Release : 2005

Latest Version : 23.11.1 (16 November 2023)

Origins : Evolved from DOS-based PCB design tool 'Protel PCB' launched in 1985

Main Functional Areas: Schematic capture, 3D PCB design, FPGA development, data management

Key Features : 3D visualization of PCB assembly, integration with component distributors

Altium 365 : Cloud-based platform connecting stakeholders in PCB design

Altium Designer is a powerful, all-in-one PCB design software that combines schematic capture, layout, and manufacturing documentation into a single intuitive interface. With its unified data model and advanced features, Altium Designer streamlines the entire design process from start to finish.

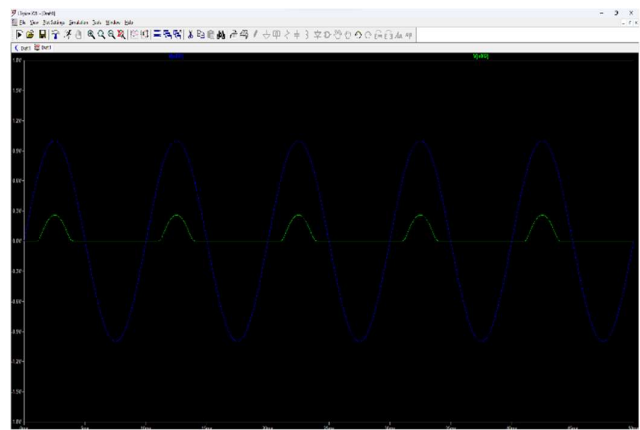
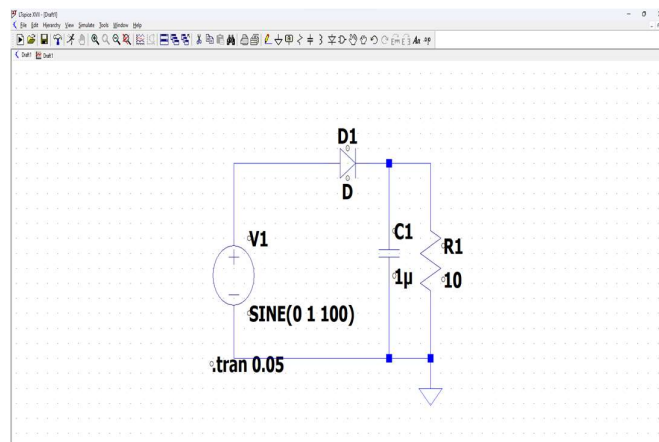
Key features of Altium Designer include:

- Precision and speed for meeting challenging design requirements and tight timelines
- Integrated SPICE simulation, power analysis, and signal integrity checks
- Support for complex designs such as HDI, high-speed, multi-board, and rigid flex
- Seamless collaboration across locations with Altium 365, enabling co-design and optimized resource allocation
- Efficient routing of high-speed electronics and dense layouts at any angle with an industry-leading interactive routing engine
- Accurate design and visualization of HDI structures for improved performance
- Secure, version-controlled data management and real-time component information with Altium 365, reducing risks and enhancing manufacturability decisions

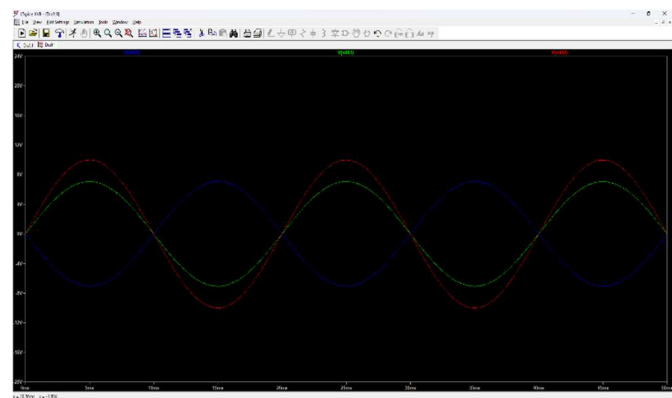
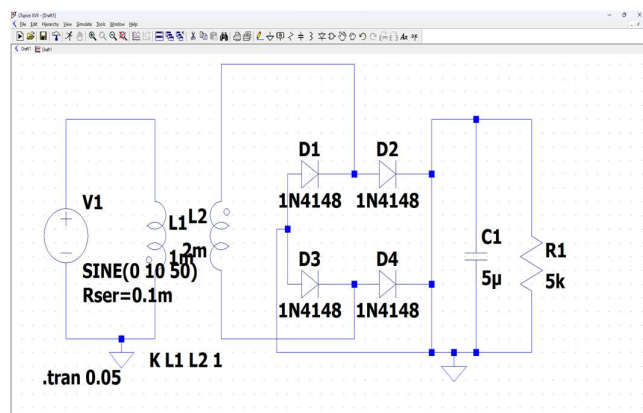
Chapter – 3

Fundamentals of Power electronics simulations in LT spice

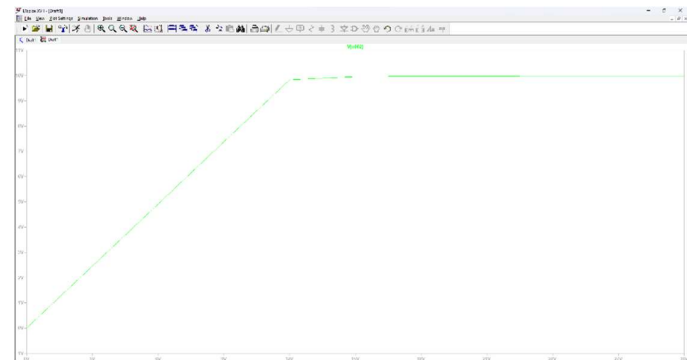
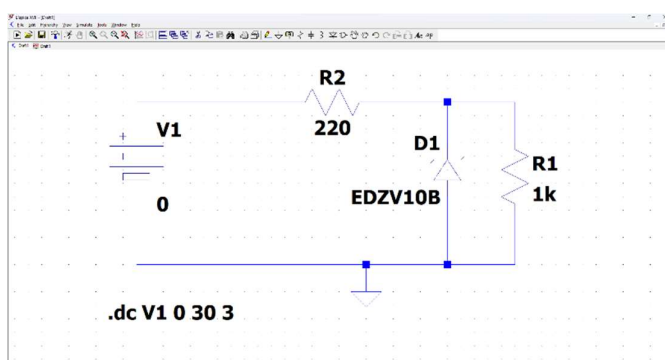
Half-Wave Rectifier



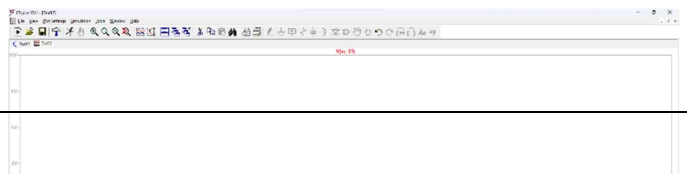
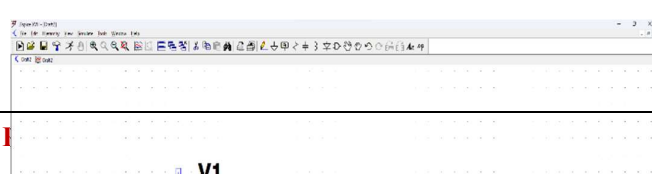
Full wave Rectifier

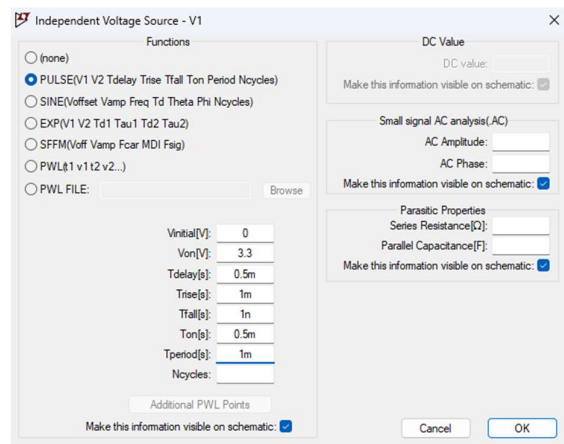


Zener Diode



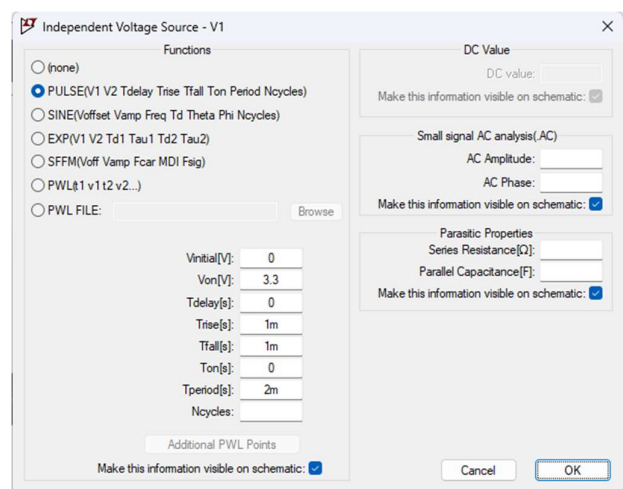
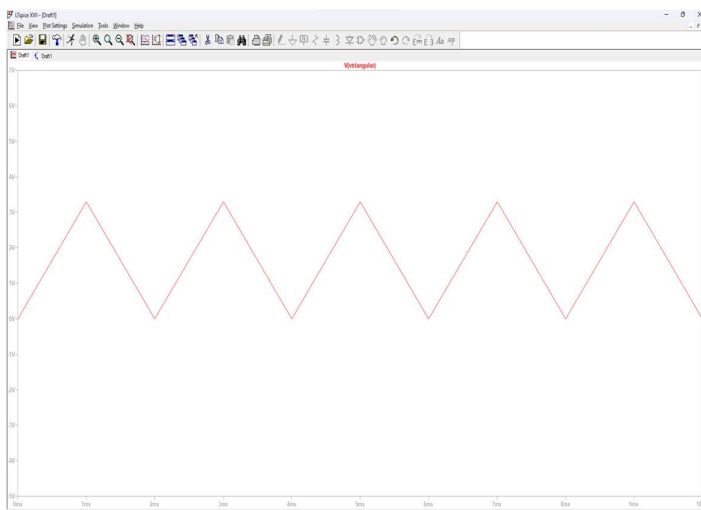
Pulse generation Voltage (Voltage source)



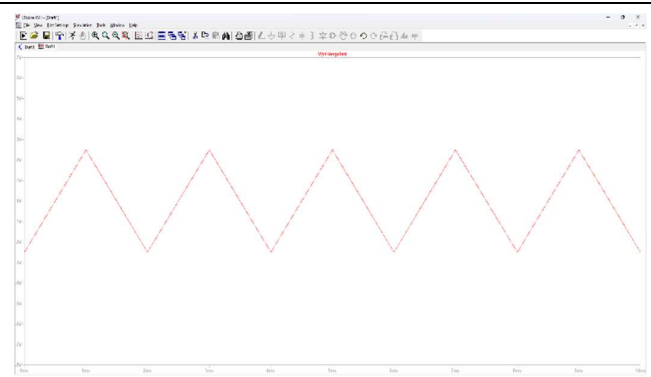
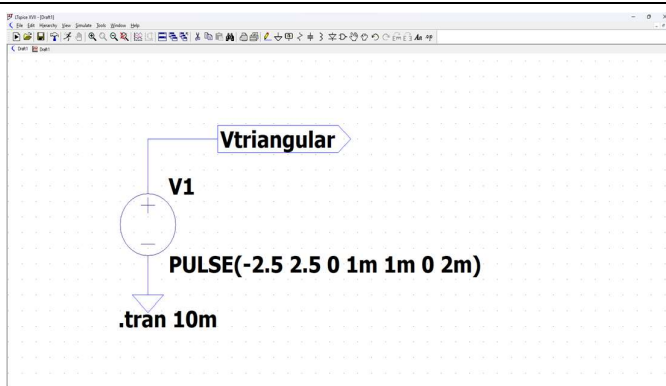


Pulse Generation – Triangular using Voltage source

Half Cycle



Full Cycle



Independent Voltage Source - V1

Functions

- ☐ (none)
- ☒ PULSE(V1 V2 Tdelay Trise Tfall Ton Period Ncycles)
- ☐ SINE(Voffset Vamp Freq Td Theta Phi Ncycles)
- ☐ EXP(V1 V2 Td1 Tau1 Td2 Tau2)
- ☐ SFFM(Voff Vamp Fcar MDI Fsig)
- ☐ PWL#1 v1 t2 v2...
- ☐ PWL FILE: Browse

DC Value

DC value:

Make this information visible on schematic: ☒

Small signal AC analysis (AC)

AC Amplitude:

AC Phase:

Make this information visible on schematic: ☒

Parasitic Properties

Series Resistance[Ω]:

Parallel Capacitance[F]:

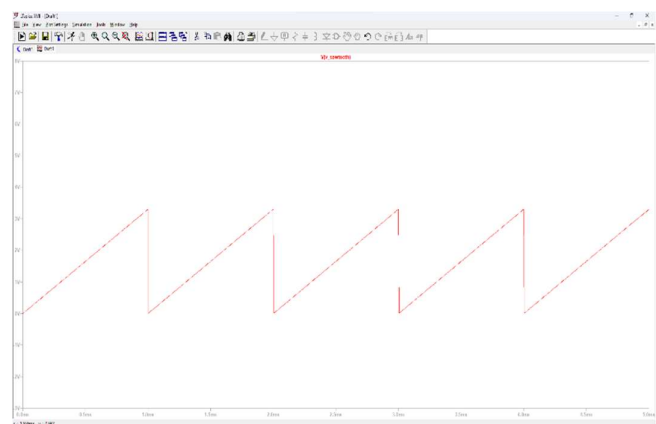
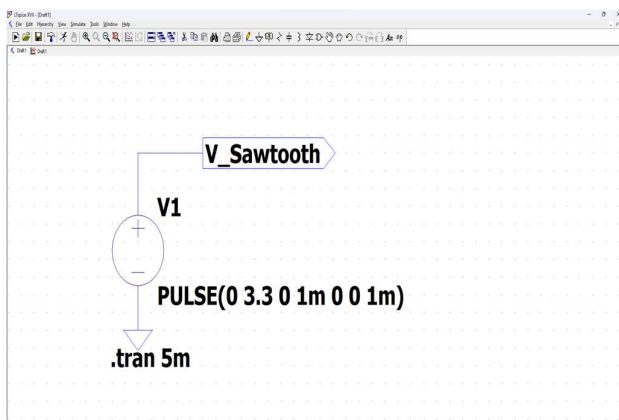
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Additional PWL Points

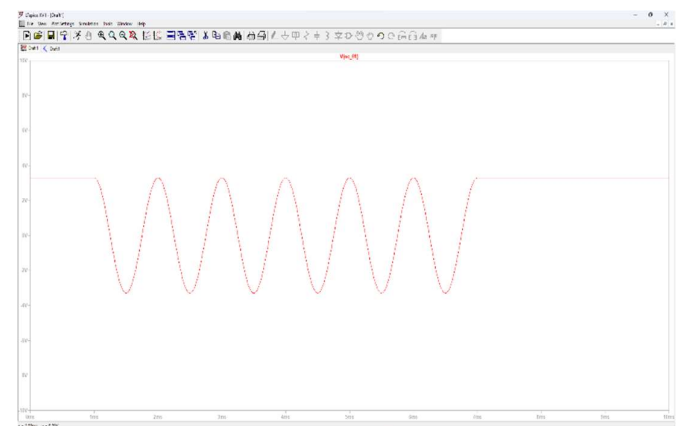
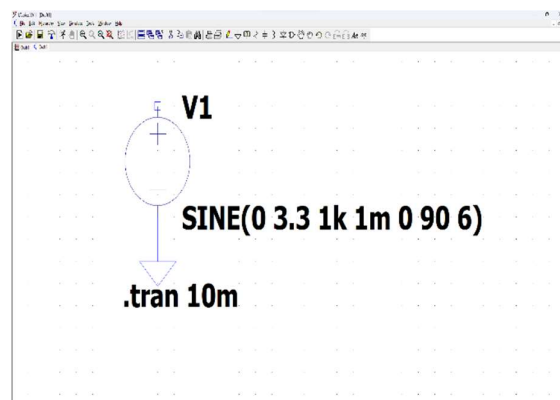
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Cancel OK

Pulse Generation – Sawtooth using Voltage source



Pulse Generation using sign wave



Independent Voltage Source - V1

Functions

- ☐ (none)
- ☒ PULSE(V1 V2 Tdelay Trise Tfall Ton Period Ncycles)
- ☐ SINE(Voffset Vamp Freq Td Theta Phi Ncycles)
- ☐ EXP(V1 V2 Td1 Tau1 Td2 Tau2)
- ☐ SFFM(Voff Vamp Fcar MDI Fsig)
- ☐ PWL#1 v1 t2 v2...

DC Value

DC value:

Make this information visible on schematic: ☒

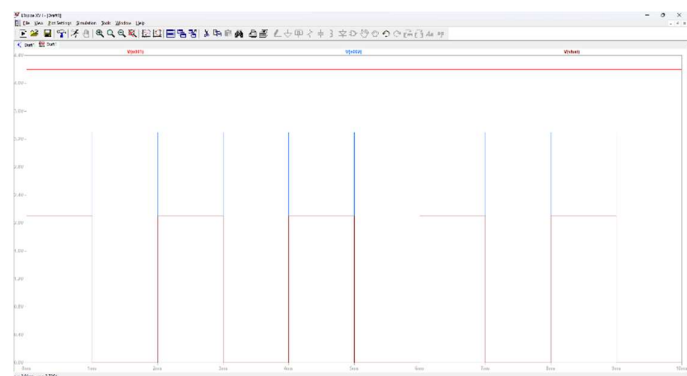
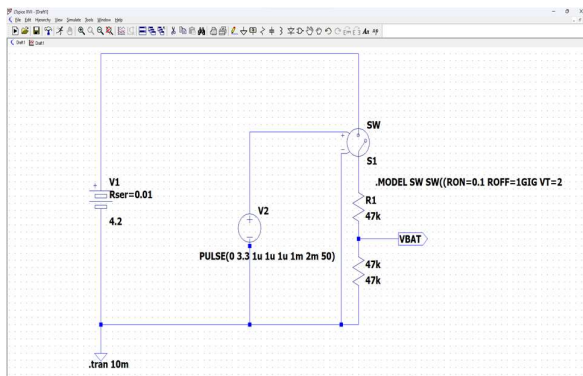
Small signal AC analysis (AC)

AC Amplitude:

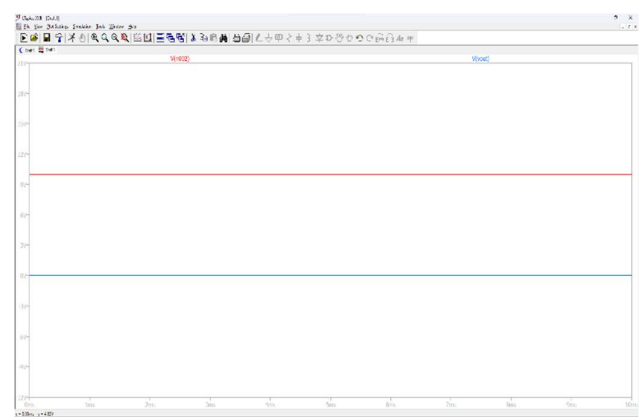
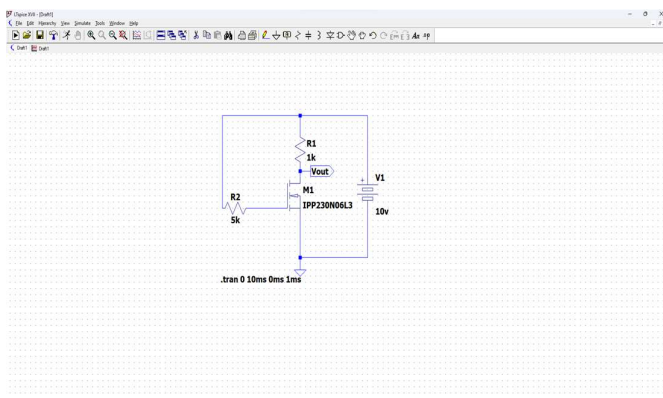
AC Phase:

Make this information visible on schematic: ☒

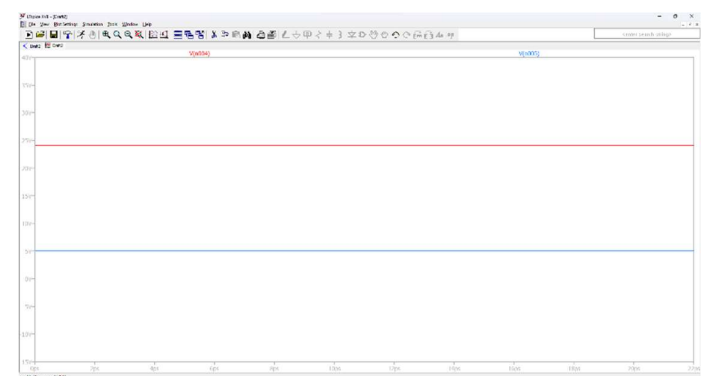
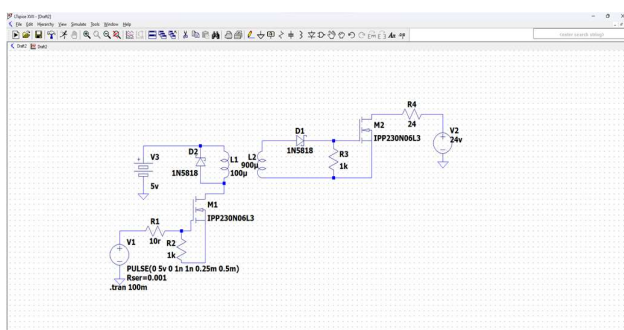
Voltage control switching



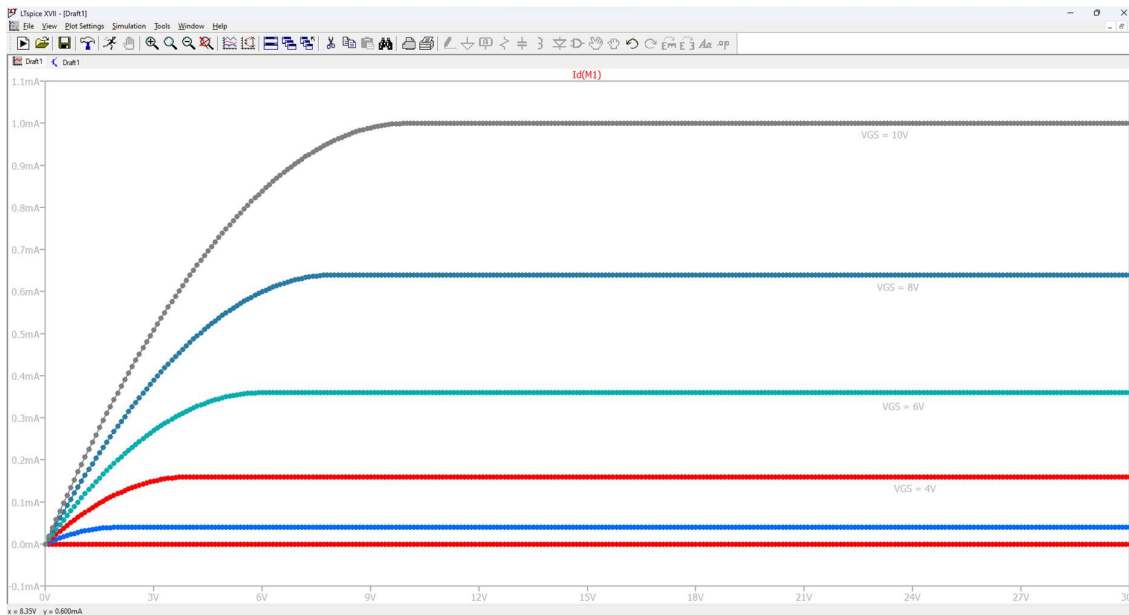
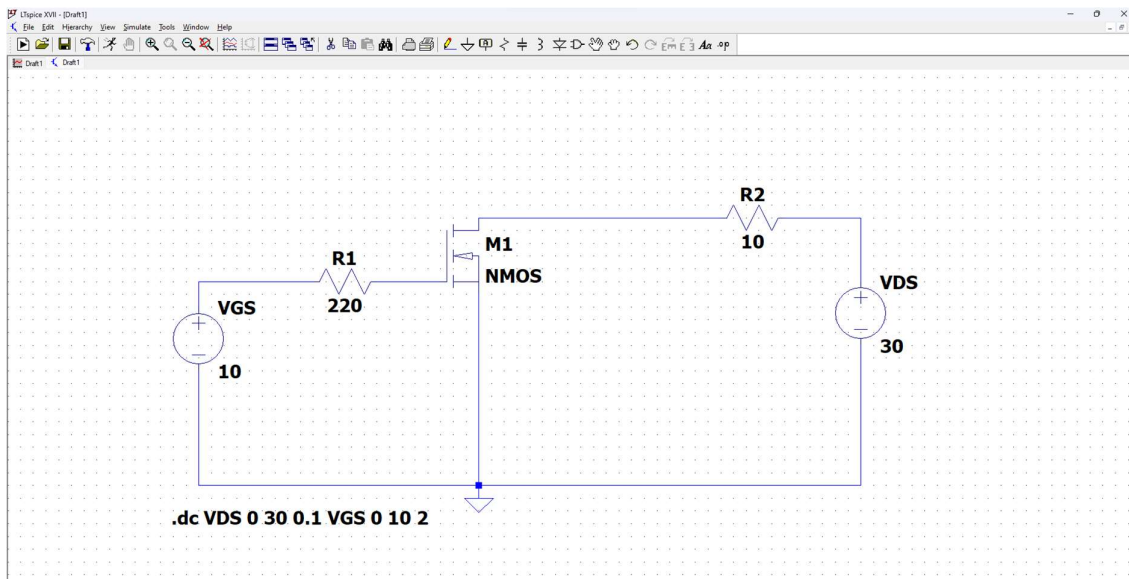
MOSFET voltage control switching



Isolated MOSFET Gate Driver using Pulse Transformer



MOSFET Characteristics

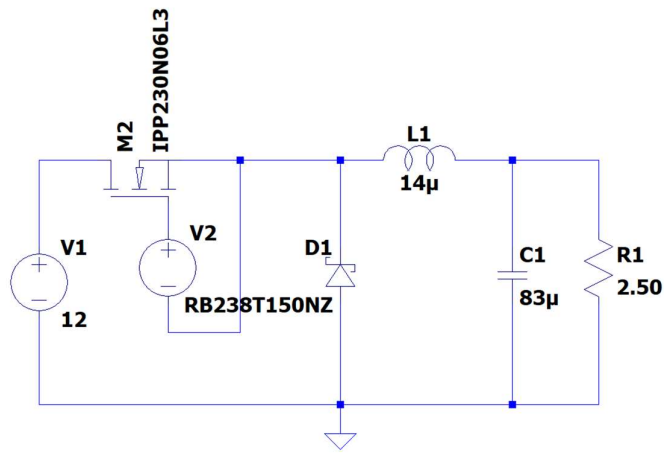


Chapter -4

Power Electronics converters

4.1. Buck Converter

A buck converter is a DC-to-DC power converter that steps down the voltage from its input (supply) to its output (load). It is a class of switched-mode power supply that uses an inductor, a controlled switch, and a diode to accomplish the voltage reduction.

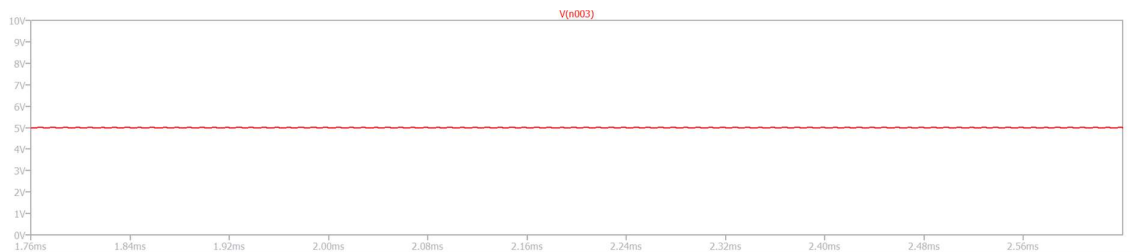


PULSE(0 12v 0 1n 1n 4.40us 10us)
.tran 1000u

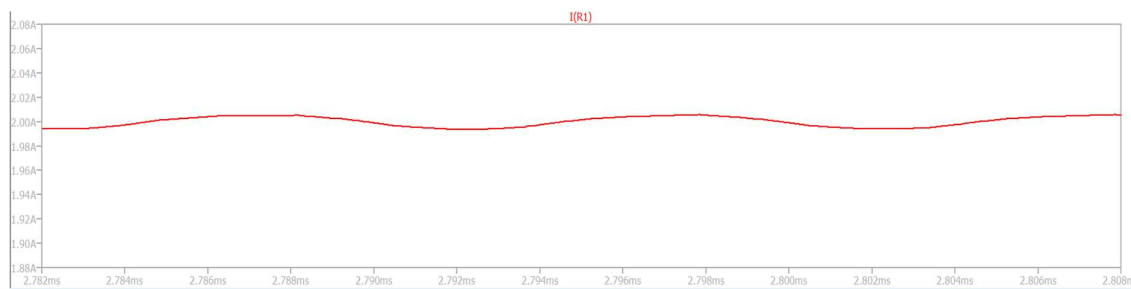
Input Voltage, $V_{in} = 12\text{ V}$
Output Voltage, $V_{out} = 5\text{ V}$
Output Power, $P_{out} = 10\text{ W}$
Switching Frequency, $F_{sw} = 100\text{ kHz}$
Inductor Ripple Current, $= 0.2\text{ A}$
Capacitance Ripple Voltage $= 0.1\text{ V}$

Duty Cycle: 0.4167
Time Period: 10μs
Output Current: 2 A
Input Current: 0.833 A
Load Resistance: 2.5 ohms
Inductance: 14.58μH
Capacitance: 83.34μF
Inductance Ripple Current: 0.2013 A
Capacitance Ripple Voltage: 0.1 mV
Diode Current: 2.2013 A

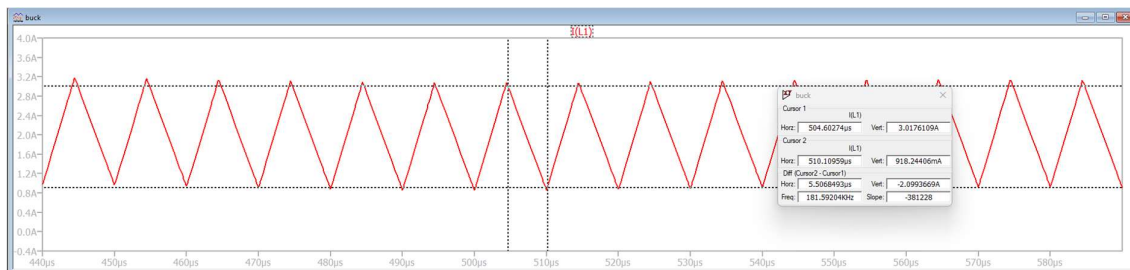
LT spice ckt design



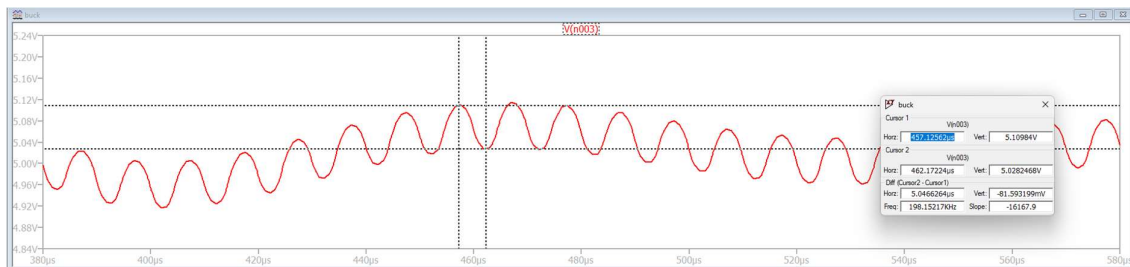
Output voltage



Output current



Inductor current ripple



Capacitor voltage ripple



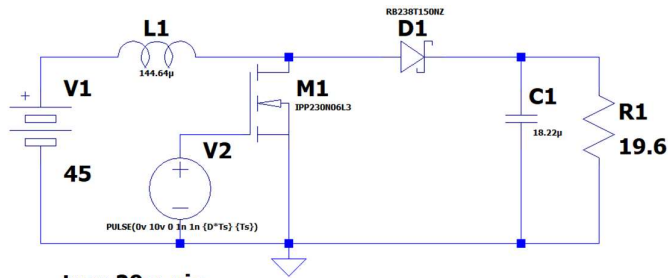
Diode Current

Buck Converter			
Input section			
Description	Symbol	Value	Unit
Minimum input voltage of converter	Vgmin	12	V
Maximum input voltage of converter	Vgmax	0	V
Nominal input voltage of converter	Vgnom		V
output voltage of converter	Vout	5	V
Switching frequency of converter	Fsw	100000	Hz
Power	P	10	W
Assumed Efficiency	Percentage	95	
Inductance Value	L	14.58	uH
Capacitance value	C	83.33	uF
Input current	I	2	A
Resistance	R	4	ohms
Assumptions			
Percentage of inductor current ripple - %	20	Percentage	
Percentage of capacitor voltage ripple - %	10	Percentage	
Calculation Section			
Capacitance voltage ripple	0.1	mV	
Capacitance value	83.33	uF	
Inductance current Ripple	0.20	mA	
Inductance value	14.58	uH	
Duty Cycle			
Min	0.417		
Max	#DIV/0!		
Frequency			
Time	100	Khz	
	10	us	
	0.01	s	
T On	4.17	us	
T off	5.83	us	
Current			
Output min	3.429	A	
Output max	2.000	A	
Input	0.877	A	
Avg input	0.833	A	
Load Resistance			
Output Voltage	5	V	
Peak-to-Peak Inductor Ripple Current	2.000	A	
Inductor RMS Current	2.082	A	
Average Diode current	1.167	A	
RMS Diode current	1.528	A	
Calculated Inductance current ripple			
Calculated Inductance current ripple	0.20	mA	
Calculated Inductance current ripple in %	20	%	
Calculated Capacitance Voltage ripple			
Calculated Capacitance Voltage ripple	0.10	mV	
Calculated Capacitance voltage ripple in %	10	%	
Calculated Power out			
Calculated Power out	10	W	
Calculated Power IN	10.53	W	
Inductor Conduction Loss (PL)			
Inductor Conduction Loss (PL)	0.400	W	
Switch Condition Loss			
Switch Condition Loss	0.0833333	W	
Total Power Loss			
Total Power Loss	1.483	W	
Calculated Efficiency			
Calculated Efficiency	95.00	%	

Calculation sheet

4.2. Boost Converter

A boost converter is a DC-to-DC converter that steps up the voltage from its input to its output. It is a class of switched-mode power supply that uses an inductor, a controlled switch, and a diode to accomplish the voltage increase

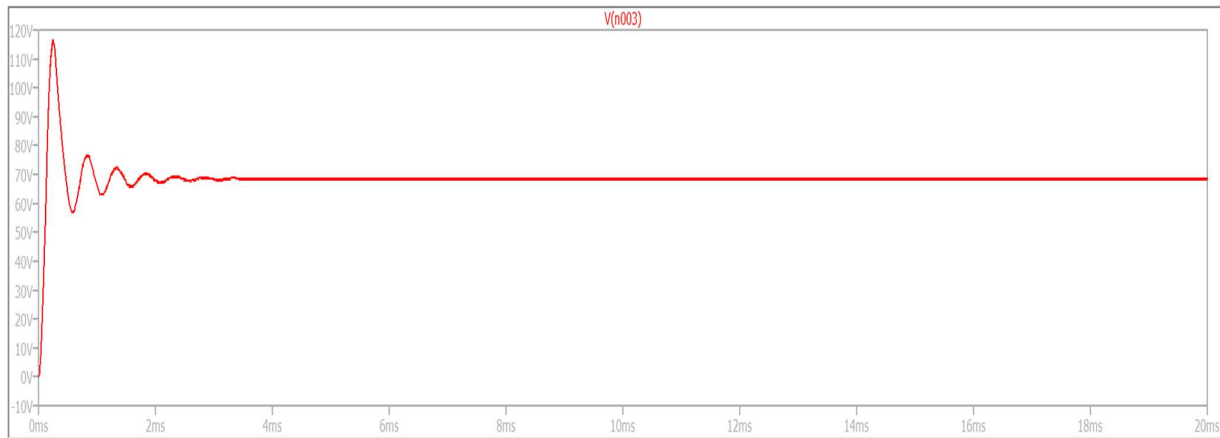


```
.tran 20m uic
.param D=0.35
.param Ts = 10u
.meas ripple_pp peak-to-peak(V(C1))
```

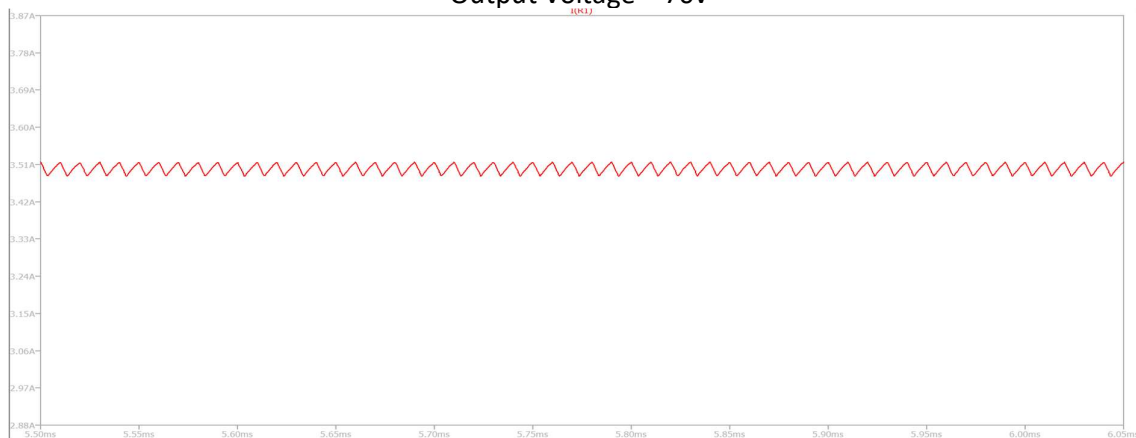
Input voltage : 45V
output voltage : 70V
Switching frequency : 100 kHz
Voltage Ripple : 0.5%
Current ripple: 10%
Power : 250W

Inductance : 144.64 uH
capacitor value : 18.22 uF
Load resistance (R): 19.6 ohms
Capacitance voltage ripple : 0.35v
Inductance current Ripple : 0.55A
Output current : 3.571A

LT Spice Circuit

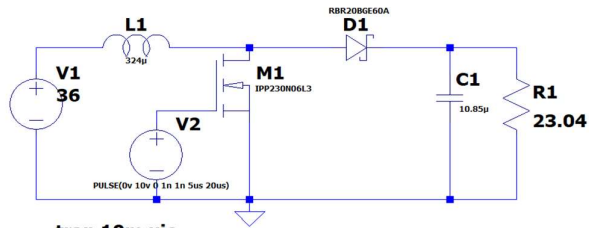


Output Voltage = 70v



Output Current = 3.57A

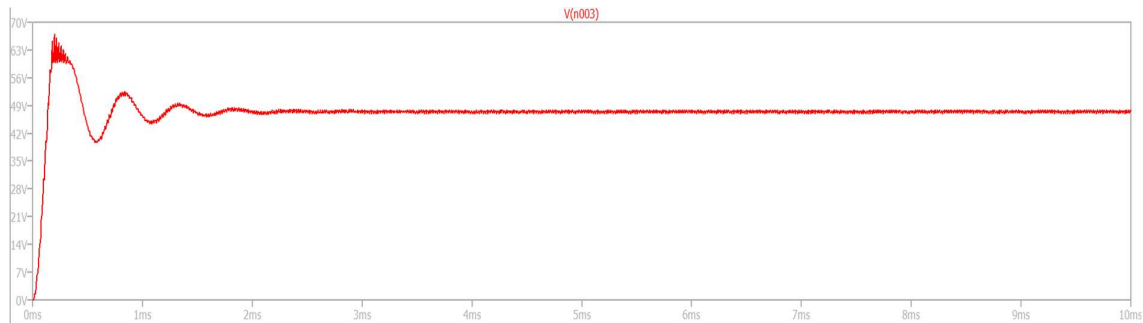
2.



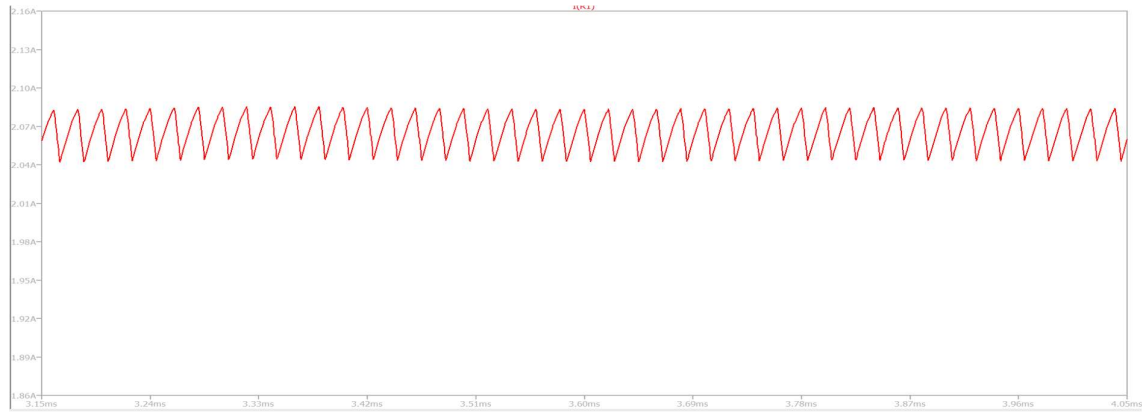
.tran 10m uic
 .param D=0.25
 .param Ts = 20u

Vin min = 12V
 Vin max = 36V
 Vout = 48V
 Power = 100W
 Frequency = 50kHz

Current = 2.083
 Calculated load resistance = 23.04 ohms.
 Duty cycle = 0.25
 Inductance L = 324uH
 Capacitance C = 10.85uF
 Calculated inductor Ripple = 0.278A
 Calculated Capacitor Ripple = 0.24V



Output Voltage = 48V



Output Current = 2.08A

Boost Converter			
Input section			
Description	Symbol	Value	Unit
Minimum input voltage of converter	Vgmin	36	V
Maximum input voltage of converter	Vgmax	68	V
Nominal input voltage of converter	Vgnom		V
output voltage of converter	Vout	48	V
Switching frequency of converter	Fsw	50000	Hz
Power	P	100	W
Assumed Efficiency	Percentage	90	
Inductance Value	L	170.5	uH
Capacitance value	C	16.2	uF
Input current	I	3.09	A
Resistace	R	23.04	ohms
Percentage of peak to peak inductor current ripple		10	Percentage
Percentage of peak to peak capacitor voltage ripple		0.5	Percentage
Calculation Section			
Capacitance voltage ripple		0.24	mV
Capacitance value		10.85	uF
Inductance current Ripple		0.278	mA
Inductance value		324.00	uH
Duty Cycle			
Min		0.25	
Max		-0.417	
Frequency			
50		Khz	
Time			
0.02		s	
T On			
5.00		us	
T off			
15.00		us	
Current			
Output min		2.778	
Output max		2.083	
Avg input		2.778	
Load Resistance			
23.04		Ohms	
Output Voltage			
48		V	
Average Inductance Current			
2.778		A	
Peak-to-Peak Inductor Ripple Current			
0.556		A	
Inductor RMS Current			
2.782		A	
Validated inductance current ripple			
1.06		A	
Validated inductance current ripple in %			
50.7		%	
Validated Capacitance Voltage ripple			
0.643004		V	
Validated Capacitance voltage ripple in %			
64		%	
Calculated Power OUT			
100		W	
Calculated Power IN			
111.11		W	
Inductor Conduction Loss (PL)			
0.772		W	
Switch Condition Loss			
0.0542535		W	
Total Power Loss			
1.826		W	
Calculated Efficiency			
90.00		%	
Voltage pluse			
Voff	0	v	
Von MOSFET	10V	v	
T delay	0	s	
T rise MOSFET	10	ns	
Tfall MOSFET	10	ns	
T on	5.00	us	
T period	20	us	
Tstop (Steady state)	5.00	ms	
Transient	20	ms	

Calculation Sheet

4.3. Buck-Boost Converter

A buck-boost converter is a type of DC-to-DC converter that has an output voltage magnitude that is either greater than or less than the input voltage magnitude. It is a combination of a buck converter and a boost converter.

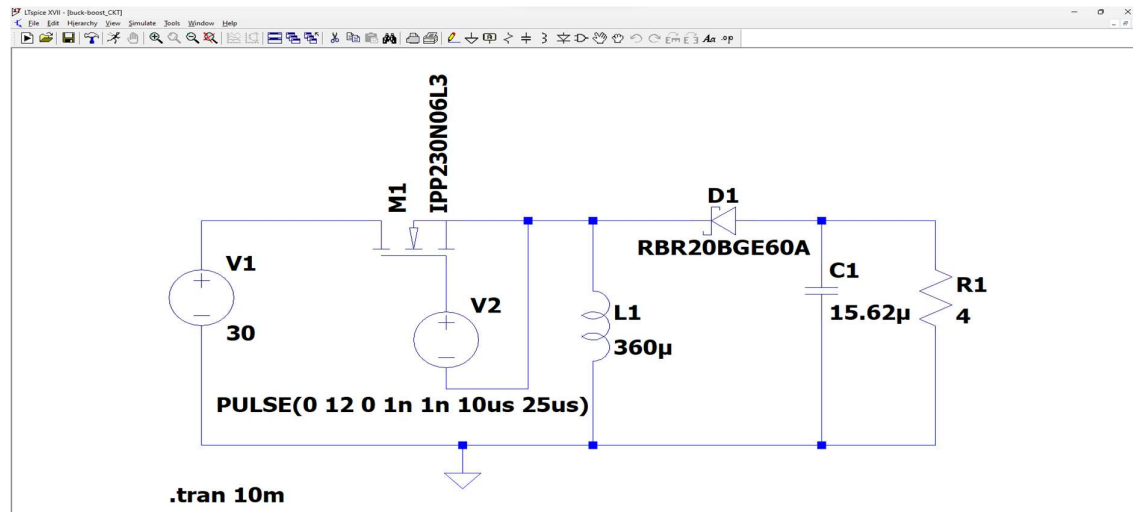
Given data

$V_{in} = 30 \text{ V}$

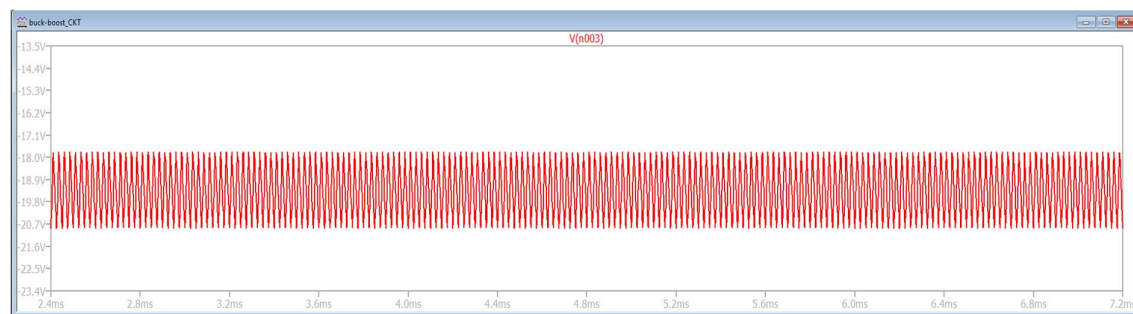
$R = 4 \Omega$

$V_{out} = -20$

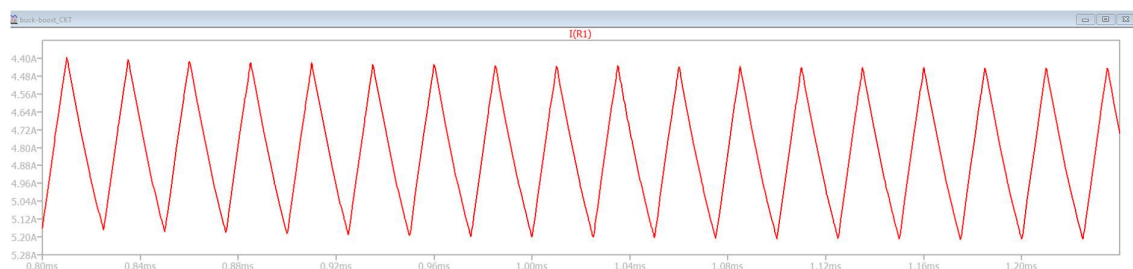
$V_f = 40 \text{ kHz}$



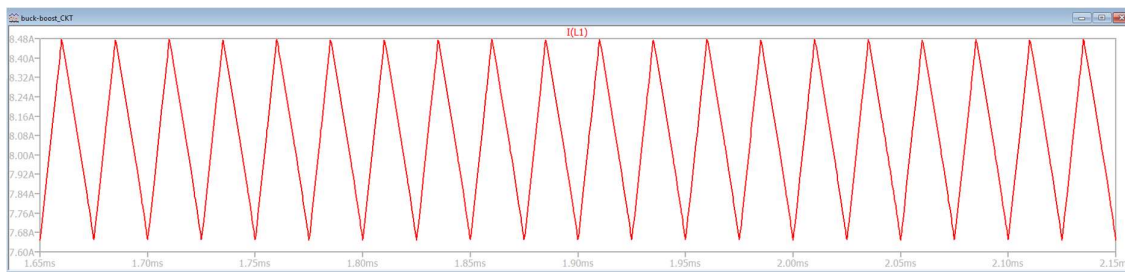
LT Spice Simulation



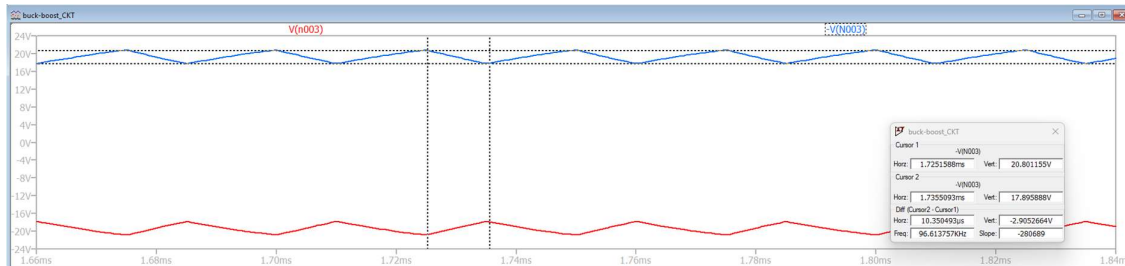
Output voltage



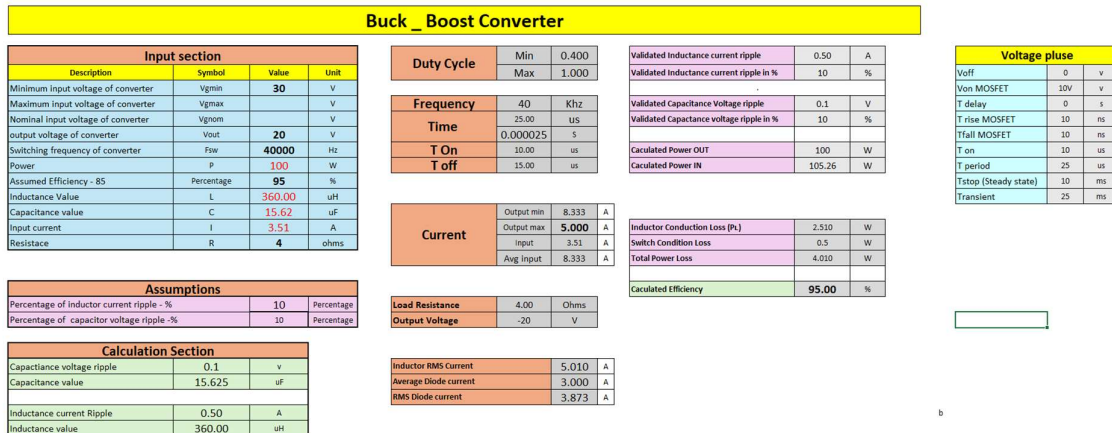
Output current



Inductance current ripple



Capacitance Voltage ripple



4.4. Flyback Converter

A flyback converter is a type of isolated buck-boost DC-to-DC converter. It has an advantage over the regular buck-boost converter in that it provides isolation between the input and output. The transformer provides the isolation and the ability to step-up or step-down the voltage

Figure 2 shows the schematic of the Flyback converter operated at a switching frequency of 20 kHz and duty cycle of 0.4. Assume all components are ideal. Sketch the following waveforms and indicate appropriate values [current minimum, current maximum and voltage magnitudes]. For calculations, you may assume 10% current ripple on the primary side of transformer

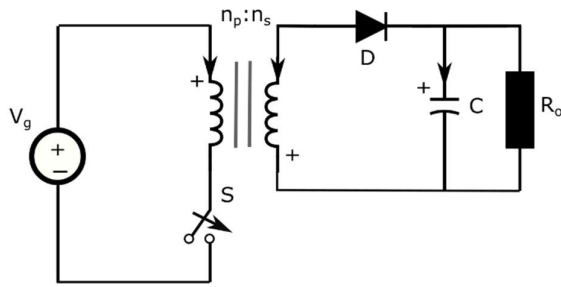
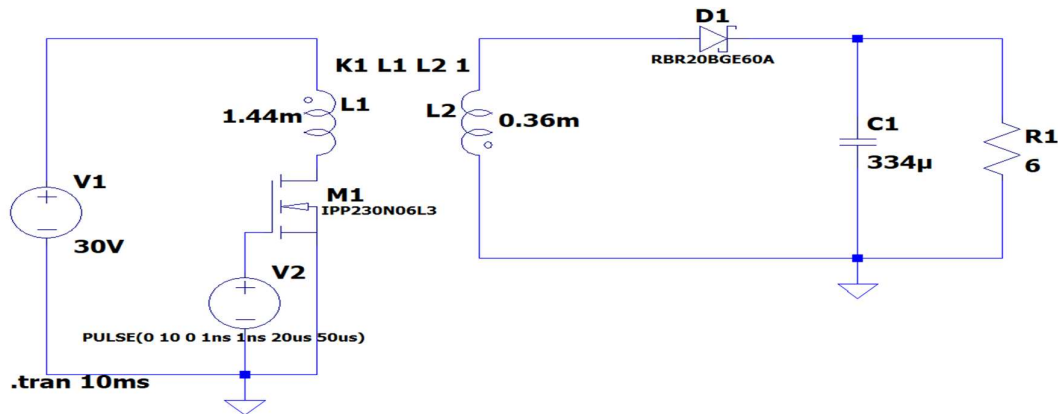
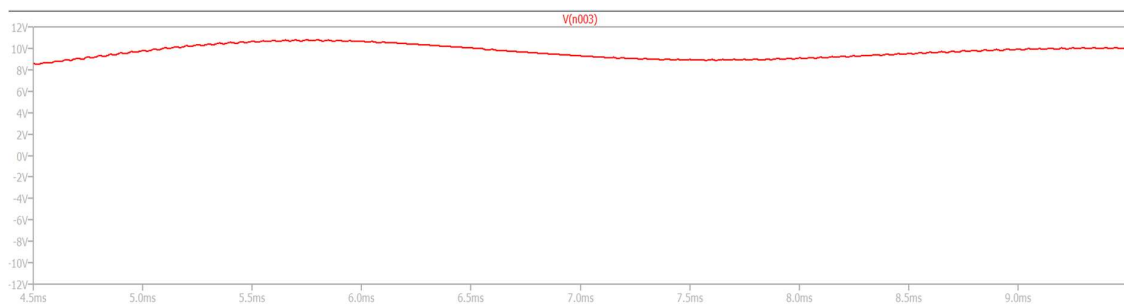


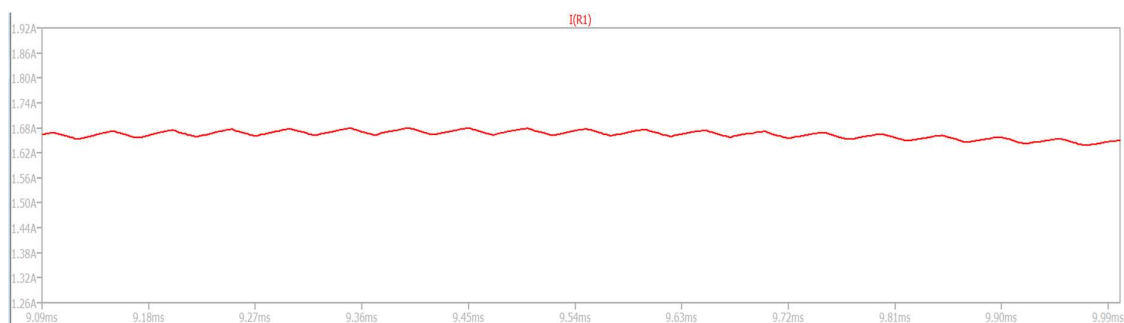
Fig. : A Flyback converter with $V_g = 30$ V, $n_p:n_s = 1:0.5$, $R_o = 6$ Ohms [plus signs at the winding are similar to your dot notations].



LT spice Simulation

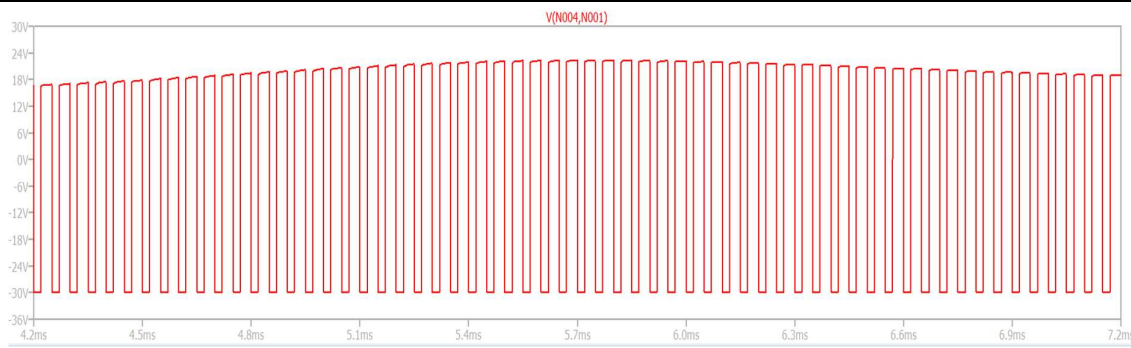


Output voltage = 8.5v

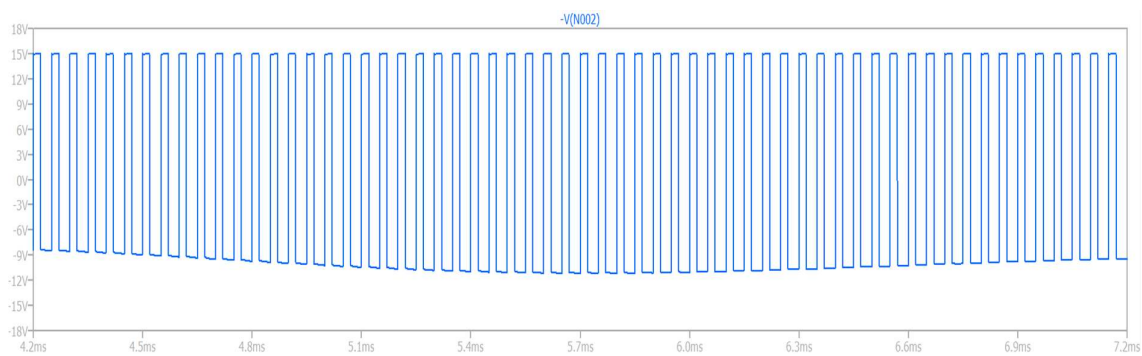


Output current = 1.65A

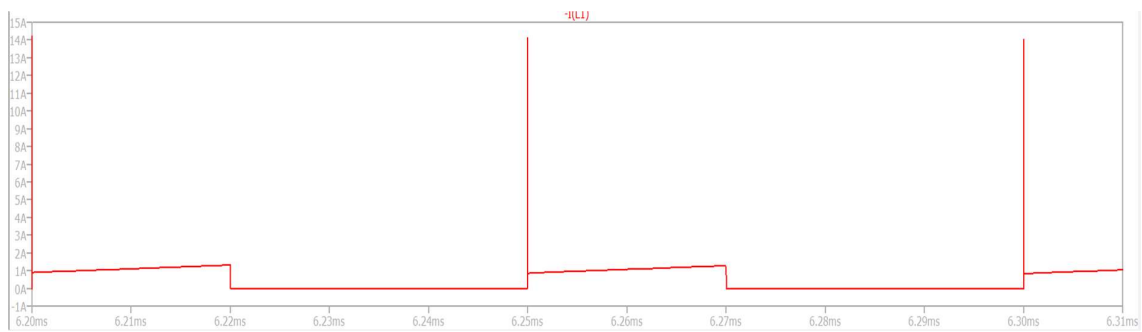
1. Primary voltage waveforms



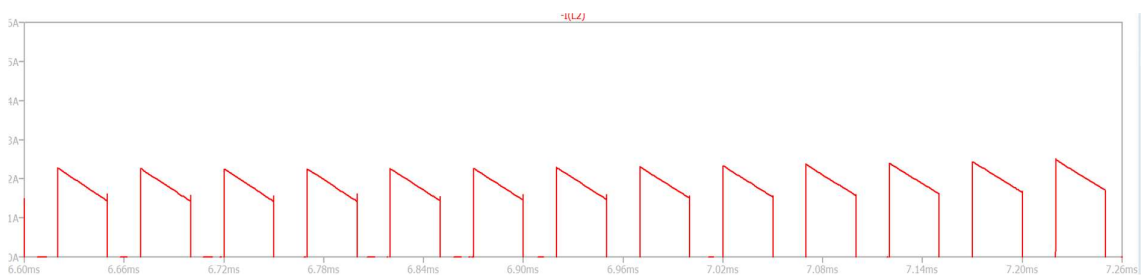
secondary voltage waveforms



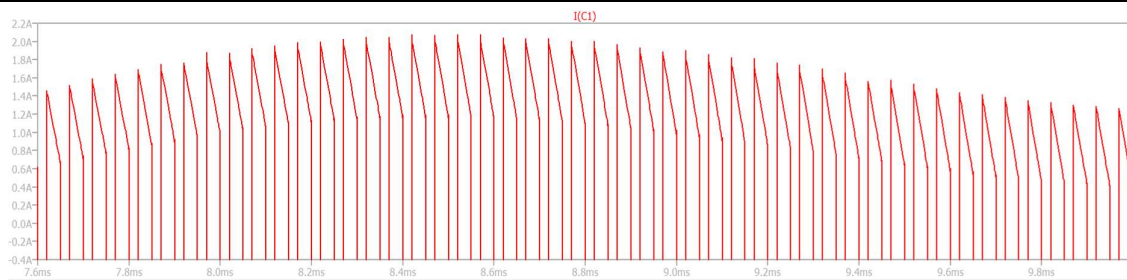
Primary current waveforms



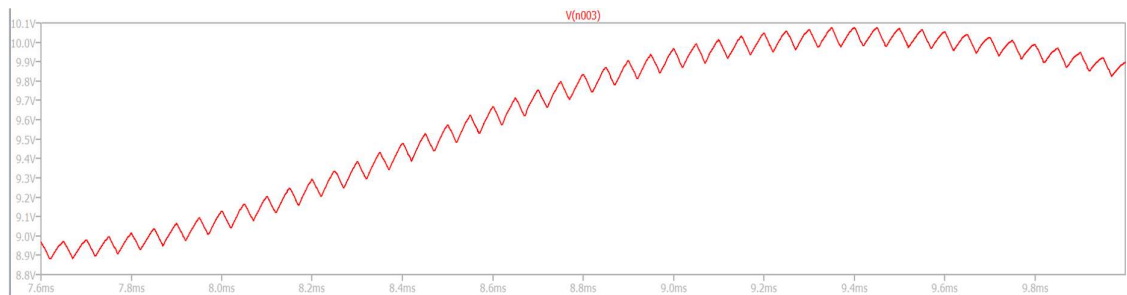
secondary current waveforms



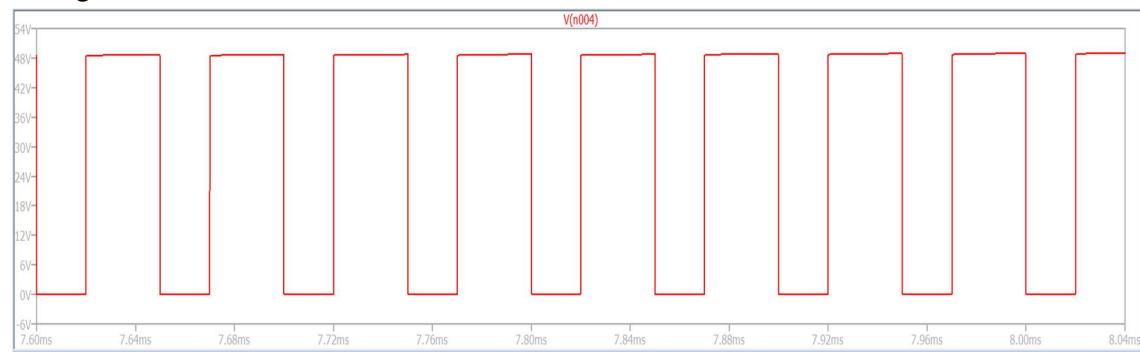
Capacitor current



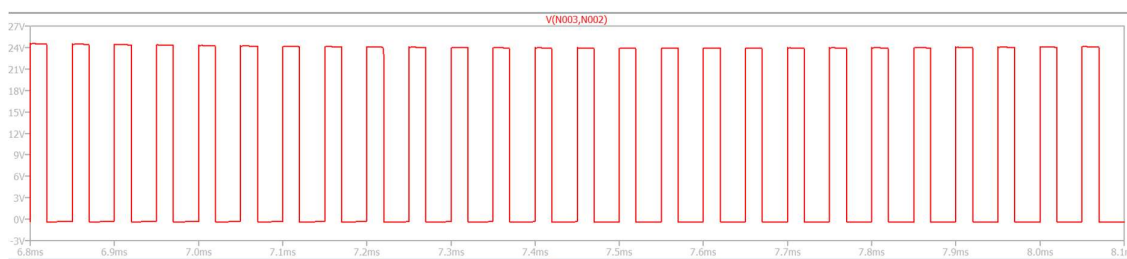
Capacitor voltage



Voltage waveforms of switch and diode



MOSFET [IPP230N06L3] Voltage waveforms



Diode [RBR20BGE60A] Voltage waveforms

FLYBACK_Covertor_DCM_CKT

Input section			
Description	Symbol	Value	Unit
Minimum input voltage of converter	Vgmin	30	V
Maximum input voltage of converter	Vgmax		V
Nominal input voltage of converter	Vgnom		V
output voltage of converter	Vout	10	V
Switching frequency of converter	Fsw	20000	Hz
Power	P	17	W
Turns Ratio	Np	1	
	Ns	0.5	
Assumed Efficiency - 85	Percentage	95	%
Inductance Value	L	360.00	uH
Capacitance value	C	15.62	uF
Input current	I	0.58	A
Resistace	R	6	ohms
Primary Inductance Duty cycle	DC	0.4	

Assumptions			
Percentage of capacitor voltage ripple - %	10	Percentage	
percentage of Inductance ripple current - %	10	Percentage	

Calculation Section		
Capactiance voltage ripple	0.1	v
Capacitance value	0.333	uF

Duty Cycle	Min	0.143
	Max	1.000

Frequency	20	KHz
	Time	
	50.00	us
	0.00005	s
T On	7.14	us
T off	42.86	us

Current	Output min	1.944	A
	Output max	1.667	A
	input	0.58	A
	Avg input	0.585	A

Load Resistance	5.88	Ohms
Output Voltage	10	V

RMS Switch Inductance	0.241	A
Inductor RMS Current	1.291	A
Average Diode current	1.429	A
RMS Diode current	1.543	A

Calculated Power OUT	17	W
Calculated Power IN	17.54	W

Inductor Conduction Loss (PL)	0.006	W
Switch Condition Loss	0.01984127	W
Total Power Loss	1.026	W
Calculated Efficiency	95.00	%

Inductor Current	4.167	A
Inductance Ripple current	0.4167	A
Average switch Inductance	0.234	A
Primary inductance Ripple	0.0417	A

Primary Inductance - Lp	1.44	mH
Secondary Inductance - Ls	0.36	mH

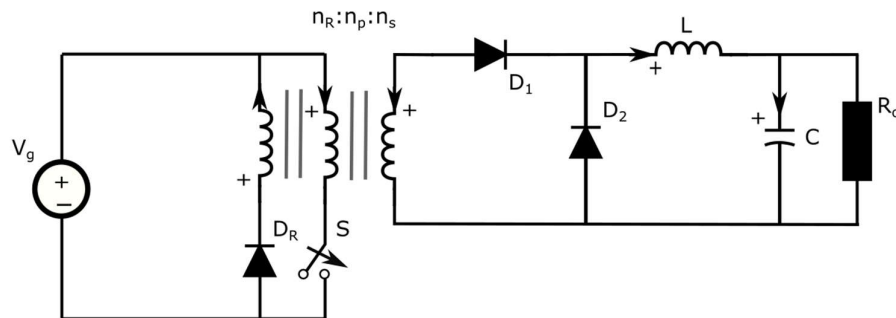
Blocking Voltage of the switch	35	V
Blocking voltage of the Diode	15	V

Voltage pluse		
Voff	0	v
Von MOSFET	10V	v
T delay	0	s
T rise MOSFET	10	ns
Tfall MOSFET	10	ns
T on	7.14	us
T period	50	us
Tstop (Steady state)	7.14	ms
Transient	50	ms

4.5. Forward Converter

A forward converter is a type of DC-to-DC converter that transfers energy from the primary to the secondary winding of a transformer while the switching element is in the "on" state. It is commonly used to either increase or decrease DC voltages or convert a single voltage to multiple DC output voltages.

Figure shows the schematic of the Forward converter operated at a switching frequency of 20 kHz and duty cycle of 0.4. Assume all components are ideal. Sketch the following waveforms and indicate appropriate values [current minimum, current maximum and voltage magnitudes].



Switching Frequency : 20 kHz

Duty Cycle (D): 0.4

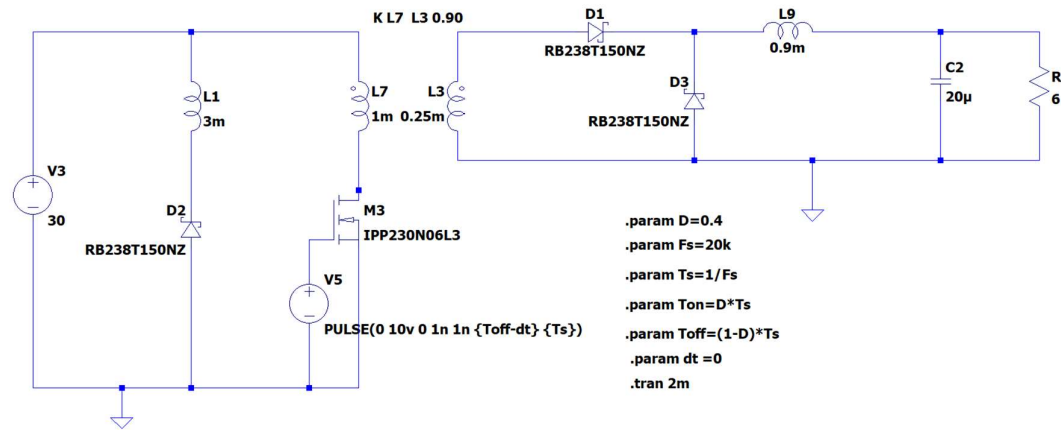
Input Voltage : 30 V

Transformer Turns Ratio : 1:1:0.5

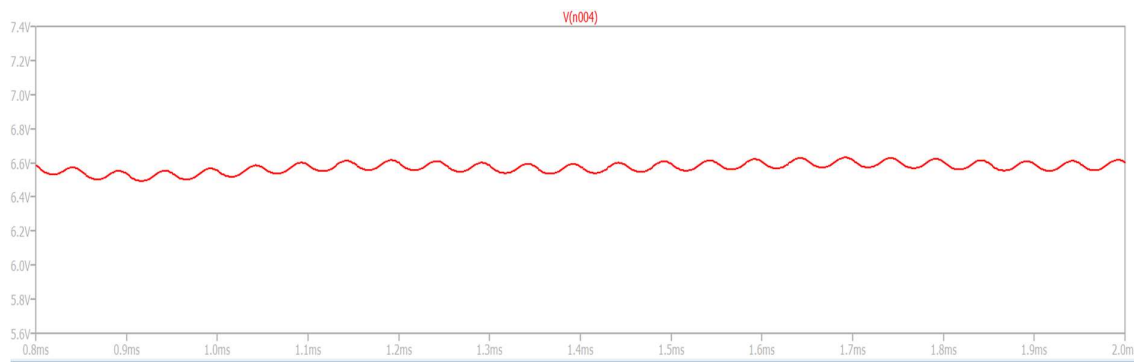
Load Resistance : 6 Ω

Inductor : 1 mH

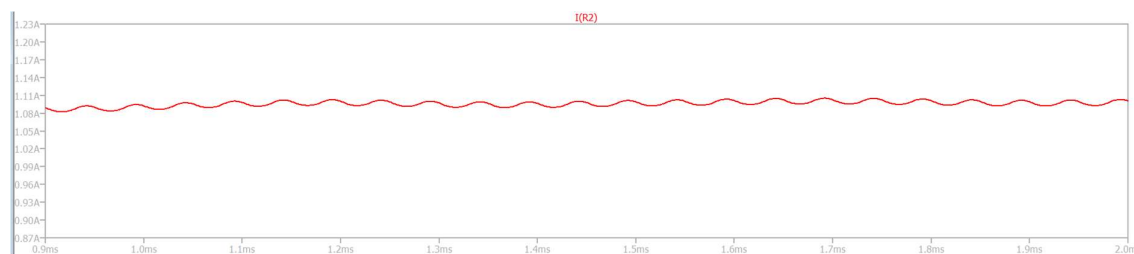
Capacitor : 20 μF



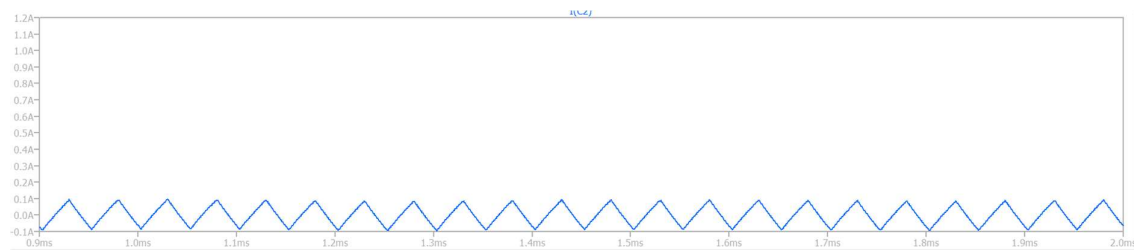
LT Spice Circuit



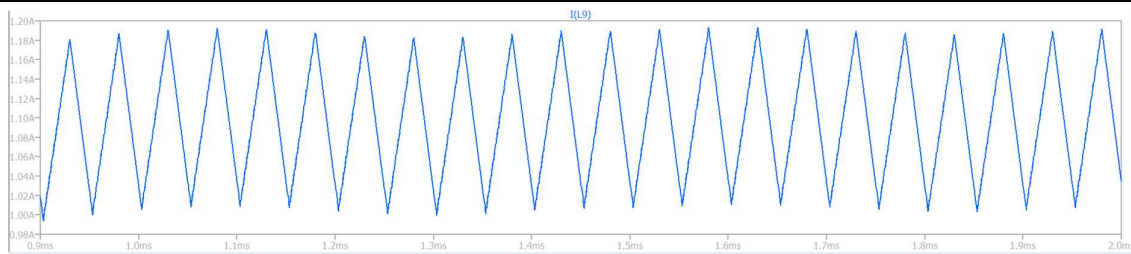
Output Voltage



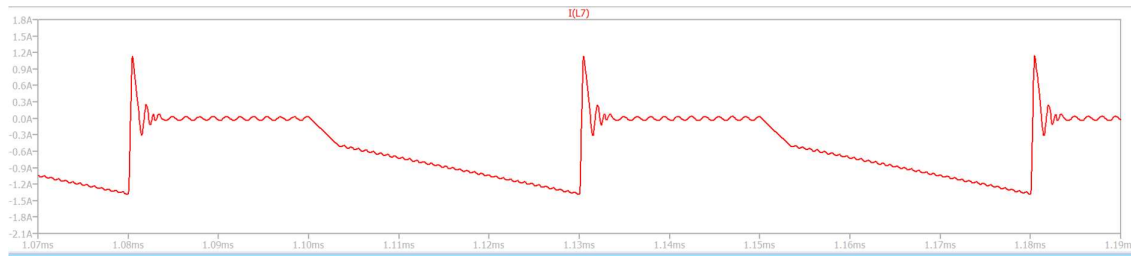
Output current



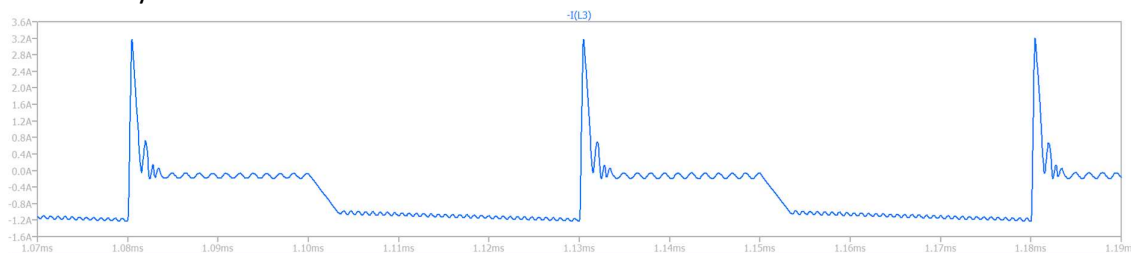
Capacitor Current



Inductance Current



Primary Inductance Current



Secondary inductance current

Forward_Coverter_Formula_sheet

Input section			
Description	Symbol	Value	Unit
Minimum input voltage of converter	Vgmin	30	V
Maximum input voltage of converter	Vgmax		V
Nominal input voltage of converter	Vgnom		V
output voltage of converter	Vout	6	V
Switching frequency of converter	Fsw	20000	Hz
Power	P	6	W
Turns Ratio	Np	1	
	Ns	0.5	
Assumed Efficiency - 85	Percentage	95	%
Inductance Value L1	L	1.00	mH
Capacitance value	C	20.00	uF
Input current	I		A
Resistance	R	6	ohms
Primary Inductance Duty cycle	DC	0.4	

Assumptions		
Percentage of capacitor voltage ripple - %	10	Percentage
percentage of Inductance ripple current - %	10	Percentage

Calculation Section		
Capacitance voltage ripple	1	mV
Capacitance value	20	uF
Inductance ripple current	0.18	A
Inductance value	1	mH

Duty Cycle	Min	0.400
	Max	1.000

Frequency	20	Khz
Time	50.00	us
T On	0.00005	s
T off	20.00	us
	30.00	us

Current	Output min	1.667	A
	Output max	1.000	A
	Input	0.21	A
	Avg Input	0.211	A

Load Resistance	6.00	Ohms
Output Voltage	6	V

RMS Switch Inductance	0.087	A
Inductor RMS Current	0.775	A
Average Diode current	0.600	A
RMS Diode current	0.775	A

Calculated Power OUT	6	W
Calculated Power IN	6.32	W

Inductor Conduction Loss (PL)	0.001	W
Switch Condition Loss	0.02	W
Total Power Loss	1.021	W
Calculated Efficiency	95.00	%

Inductor Current	2.500	A
Inductance Ripple current	0.2500	A
Average switch Inductance	0.0842	A
Primary inductance Ripple	0.0250	A

Primary Inductance - Lp	2.4	mH
Secondary Inductance - Ls	0.4	mH

Blocking Voltage of the switch	33	V
Blocking voltage of the Diode	15	V

Primary current of a Transformer	2.5	A
Secondary current of a Transformer	1.25	A

Voltage pluse		
Voff	0	v
Von MOSFET	10V	v
T delay	0	s
T rise MOSFET	10	ns
Tfall MOSFET	10	ns
T on	20.00	us
T period	50	us
Tstop (Steady state)	20.00	ms
Transient	50	ms

b

4.6 push-pull.

A push-pull converter is a type of isolated DC-to-DC converter that uses two transistors connected to a centre-tapped transformer. It is a variation of the buck-boost converter. The push-pull converter can provide a step-up or step-down output voltage.

$V_{in} = 12\text{V}$

$V_{out} = 24\text{V}$

Output power = 50W

Frequency = 100kHz

Duty cycle = 0.4

Assume - Peak current = 5A

Calculate – Transformer turns ratio, output current, output voltage, primary and Secondary inductance values, output capacitor filter, output filter inductance, Load resistance?

Calculated

Transformer turns ratio – 4

output current – 2.08A

output voltage – 24V

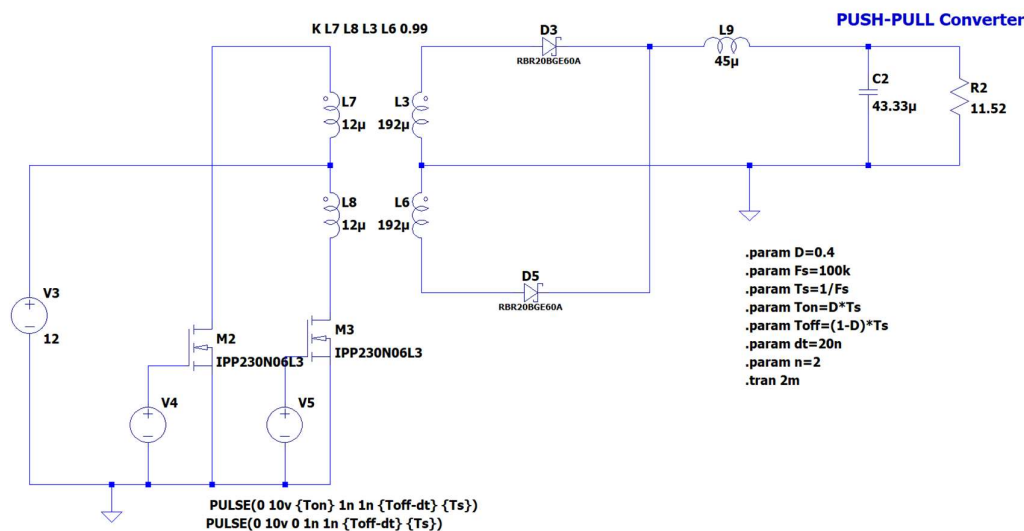
primary inductance value – 12uH

Secondary inductance value – 192uH

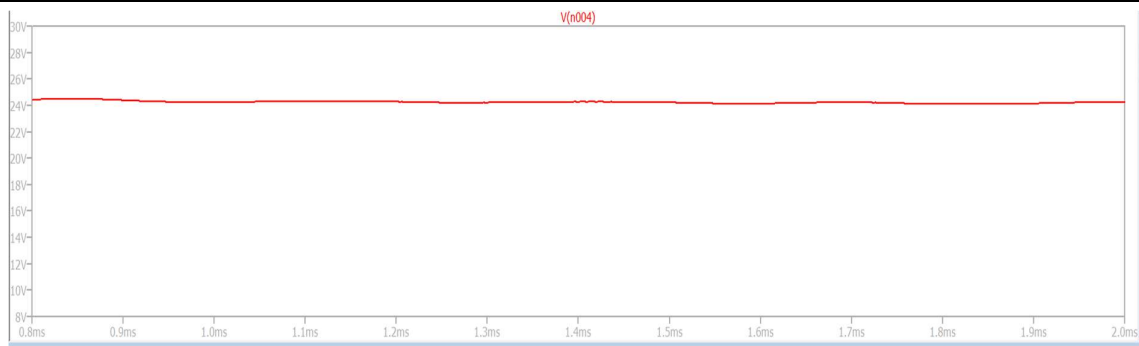
Output capacitor filter – 43.33uF

Output filter inductance – 45uH

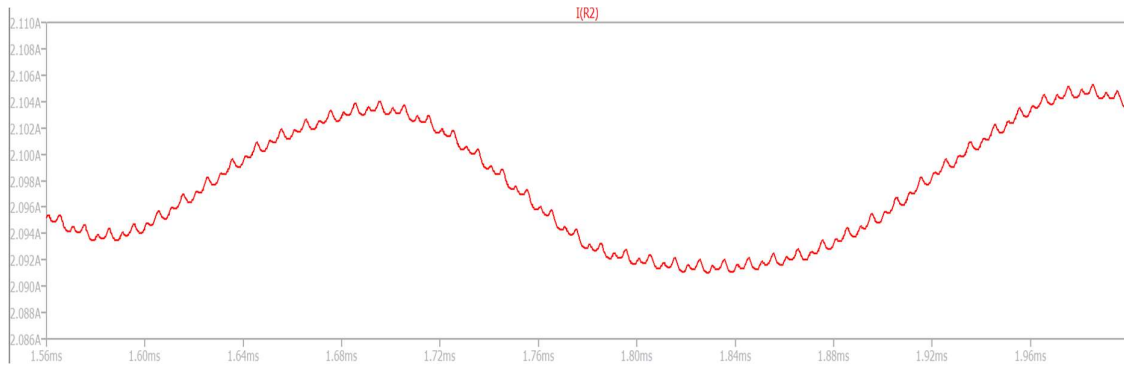
Load resistance – 11.52 ohms



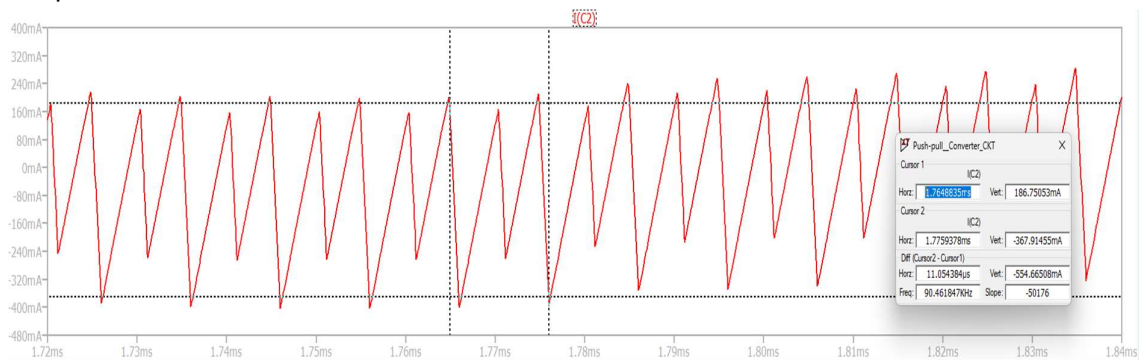
LT spice Simulation



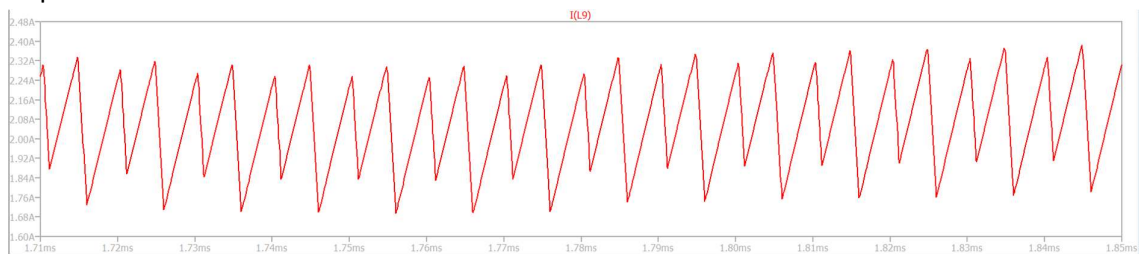
Output voltage



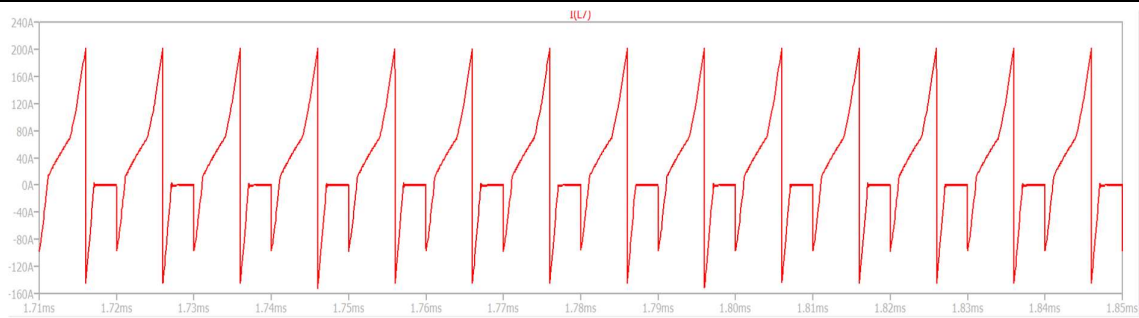
Output current



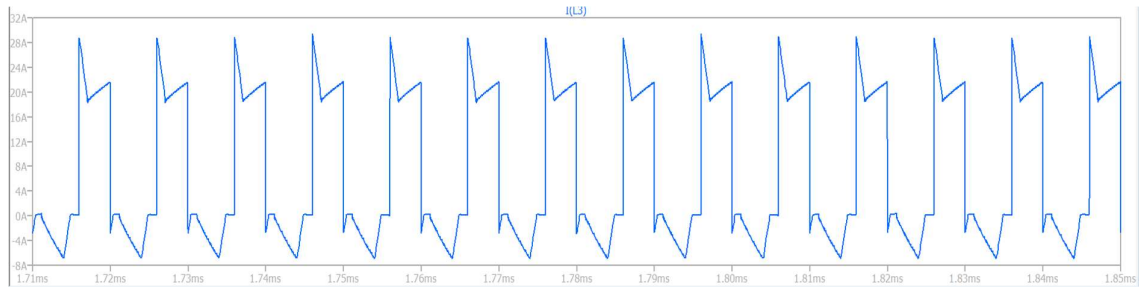
Capacitor current



Inductance current



Primary Inductance Current



Secondary Inductance current

Push-pull_Converter_Formula_sheet

Input section			
Description	Symbol	Value	Unit
Minimum input voltage of converter	Vgmin	12	V
Maximum input voltage of converter	Vgmax		V
Nominal input voltage of converter	Vgnom		V
output voltage of converter	Vout	24	V
Switching frequency of converter	Fsw	100000	Hz
Power	P	50	W
Turns Ratio	Ns	1	
	Np	4	
Assumed Efficiency - 85	Percentage	95	%
Inductance Value L1	L	45.00	mH
Capacitance value	C	43.00	uF
Input current	I	4.39	A
Resistance	R	11.52	ohms
Primary Inductance Duty cycle	DC	0.4	
Peak current		5	A

Assumptions		
Percentage of capacitor voltage ripple - %	10	Percentage
percentage of Inductance ripple current - %	20	Percentage

Calculation Section		
Capacitance voltage ripple	0.1	mV
Output Filter Voltage	43	uF
Inductance ripple current	0.2	A
Output Filter Inductance	45	mH

Duty Cycle	Min	0.500
	Max	1.000

Frequency	100	Khz
	10.00	us
Time	0.00001	s
	T On	5.00 us
T off	5.00	us

Current	Output min	4.167	A
	Output max	2.083	A
	Input	4.39	A
	Avg Input	4.386	A

Load Resistance	11.52	Ohms
Output Voltage	24	V

RMS Switch Inductance	1.802	A
Inductor RMS Current	1.614	A
Average Diode current	1.042	A
RMS Diode current	1.473	A

Calculated Power OUT	50	W
Calculated Power IN	52.63	W

Inductor Conduction Loss (PL)	0.325	W
Switch Condition Loss	0.109	W
Total Power Loss	1.433	W
Calculated Efficiency	95.00	%

Inductor Current	5.208	A
Inductance Ripple current	1.0417	A
Average switch Inductance	1.7544	A
Primary Inductance Ripple	#REF!	A

Turns Ratio	4	
Primary Inductance - Lp1 and Lp2	12	mH
Secondary Inductance - Ls1 and Ls2	192	mH

Blocking Voltage of the switch	108	V
Blocking voltage of the Diode	6	V

Primary current of a Transformer	4.16666667	A
Secondary current of a Transformer	16.66666667	A

Voltage pluse		
Voff	0	v
Von MOSFET	10V	v
T delay	0	s
T rise MOSFET	10	ns
Tfall MOSFET	10	ns
T on	5.00	us
T period	10	us
Tstop (Steady state)	5.00	ms
Transient	10	ms

Calculation sheet

Chapter -5

Gate Drivers and GaN MOSFETs

5.1. NCP51820 Gate Driver (ON Semiconductor)

NCP51820: A High-Speed Half-Bridge Driver for GaN Power Switches

The NCP51820 is a powerful and versatile gate driver designed to meet the stringent requirements of driving enhancement mode (e-mode), high electron mobility transistor (HEMT) and gate injection transistor (GIT) gallium nitride (GaN) power switches. This high-speed, half-bridge driver offers a range of advanced features that make it well-suited for a variety of power electronics applications.

Key Features

High-Speed Operation: The NCP51820 is optimized for driving GaN power switches, providing fast rise and fall times of just 1 ns. This enables efficient high frequency switching in power conversion topologies.

Wide Common Mode Voltage Range: The advanced level shift technology in the NCP51820 provides a -3.5 V to +650 V common mode voltage range for the high-side drive, and a -3.5 V to +3.5 V range for the low-side drive. This flexibility allows the driver to be used in a wide range of applications.

Stable dV/dt Operation: The NCP51820 is rated for dV/dt up to 200 V/ns, ensuring stable and reliable operation even in high-speed switching environments.

Dedicated Gate Protection: The driver employs independent voltage regulators for both the high-side and low-side drive stages. This maintains the gate-source drive signal amplitude, protecting the GaN power transistors from excessive voltage stress.

Comprehensive Protection Features: The NCP51820 includes under-voltage lockout (UVLO) monitoring for the VDD bias voltage and VDDH/VDDL driver bias, as well as thermal shutdown based on the die junction temperature.

Programmable Dead-Time Control: The dead-time between the high-side and low-side drive signals can be configured to prevent cross-conduction, ensuring safe and efficient operation.

High Drive Capability: The NCP51820 provides a source capability of 1 A and a sink capability of 2 A, enabling it to drive a wide range of GaN power switches.

Input Compatibility: The driver features dual TTL compatible Schmitt trigger inputs, simplifying integration with microcontrollers and digital signal processors.

Applications

The NCP51820 is well-suited for a variety of power electronics applications, particularly those involving soft-switching topologies:

Soft-switching full or half-bridge converters

LLC resonant converters

Active clamp flyback or forward converters

Synchronous rectifier topologies

5.2. GaN3R2-100CBE MOSFET (Nexperia)

The GAN3R2-100CBE is a 100 V, 3.2 m Ω Gallium Nitride (GaN) FET designed for high efficiency and high power density applications. It features low on-state resistance, high frequency switching capabilities, and is housed in a compact Wafer Level Chip-Scale Package (WLCSP).

Key Features:

Enhancement Mode (Normally-Off) Power Switch: Provides high efficiency and high power density.

High Frequency Switching: Capable of ultra-high frequency switching.

Low On-State Resistance ($R_{DS(on)}$): 3.2 m Ω maximum.

Low Gate and Output Charge: Reduces switching losses.

No Body Diode: Eliminates reverse recovery losses.

ESD Protection: Enhanced reliability.

RoHS, Pb-Free, REACH-Compliant: Environmentally friendly.

Applications:

Power Conversion: High power density and high efficiency applications.

AC-to-DC Converters: Particularly useful for the secondary stage.

DC-to-DC Converters: Ideal for high frequency converters in 48 V systems.

Fast Battery Charging: Suitable for mobile phones, laptops, tablets, and USB type-C chargers.

Data and Telecom: Effective in AC-to-DC and DC-to-DC converters.

Motor Drives and Class D Audio Amplifiers: High performance in these applications.

LiDAR: Suitable for non-automotive applications.

Specifications:

Drain-Source Voltage (V_{DS}): 100 V maximum.

Drain Current (I_D): 60 A maximum.

Total Power Dissipation (P_{tot}): 394 W maximum.

Junction Temperature (T_j): -40 to 150 °C.

Gate-Source Voltage (V_{GS}): -4 to 6 V.

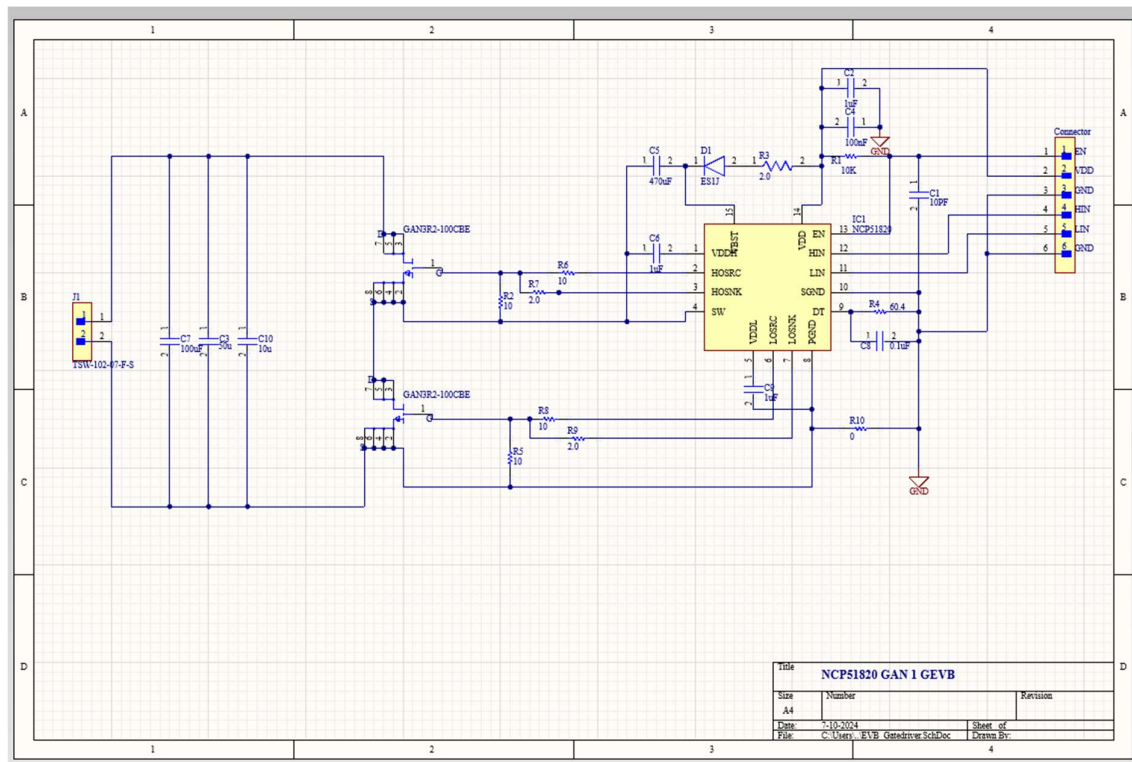
Transient Drain-Source Voltage (V_{TDS}): 130 V maximum (pulsed, $t_p = 1 \mu s$).

Gate Resistance (R_G): 2.2 Ω typical.

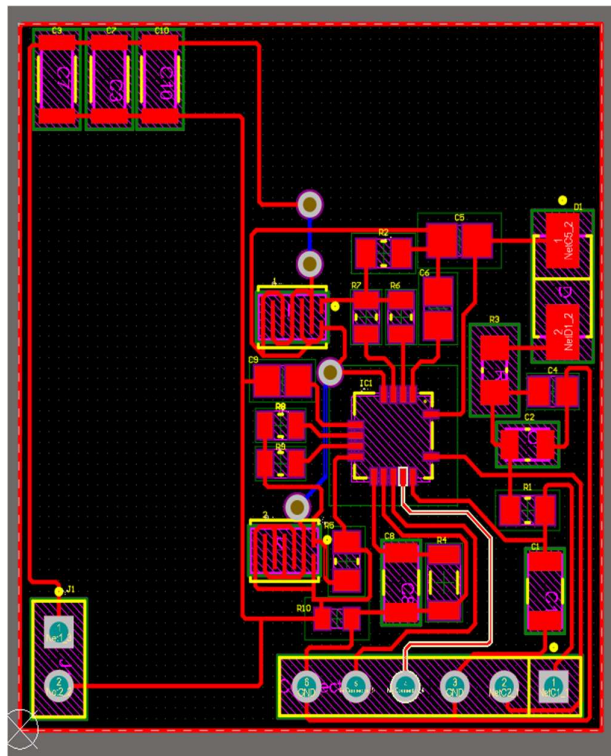
Thermal Resistance ($R_{th(j-c)}$): 0.3 K/W typical from junction to case.

Chapter – 6

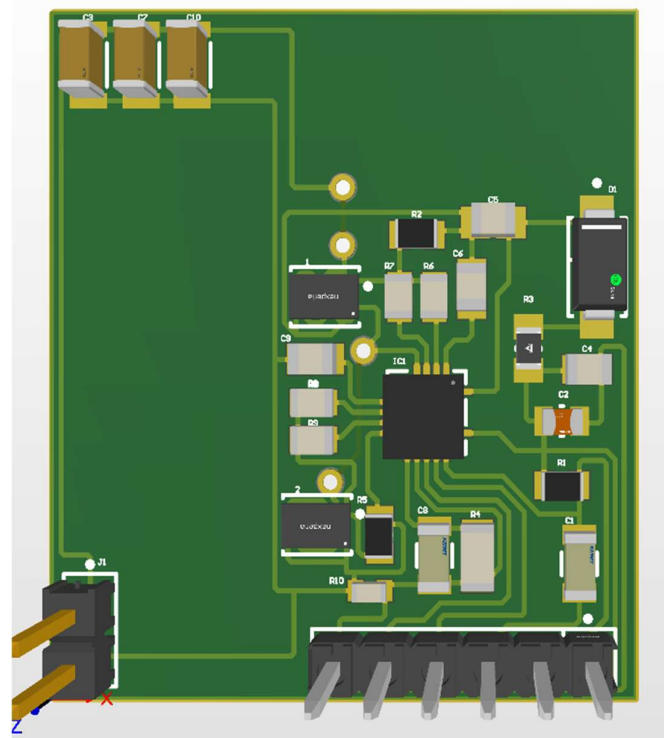
Altium Schematic and PCB Layout Design



Schematic



2D PCB Design



3D PCB Design

Conclusion

Power electronic converters, such as buck, boost, and flyback, are crucial in modern electrical engineering. They efficiently manage and convert electrical power across various applications, improving energy efficiency and reducing power losses. Buck converters step down voltages, boost converters step up voltages, and buck-boost converters can do both. Flyback converters allow multiple output voltages from a single input, and forward converters are suitable for high output currents. The choice of converter depends on the application's requirements, and careful selection is essential for optimal performance and energy efficiency.

References

Fundamentals of MOSFET and IGBT Gate Driver Circuits

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GAN3R2-100CBE

<https://assets.nexperia.com/documents/data-sheet/GAN3R2-100CBE.pdf>

High Speed Half-Bridge Driver for GaN Power Switches NCP51820

www.onsemi.com/pdf/datasheet/ncp51820-d.pdf